

# Combinatorics

**Discrete Math, Spring 2026**

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## **Graph Theory**

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- Graphs & digraphs
- Paths & connectivity
- Trees & spanning trees
- Bipartite graphs
- Matchings & Hall's theorem
- Planarity & coloring
- Network flows

## **Languages & Computation**

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- Alphabets & formal languages
- Regular expressions
- Finite automata (DFA, NFA)
- Pumping lemma
- Context-free grammars
- Pushdown automata
- Turing machines
- Decidability & complexity

## **Combinatorics & Recurrences**

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- Counting principles
- Permutations & combinations
- Inclusion-exclusion
- Partitions & Stirling numbers
- Generating functions
- Recurrence relations
- Asymptotic analysis

# Combinatorics

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*“It is better to solve one problem five different ways,  
than to solve five problems one way.”*

*— George Pólya*



George Pólya



Abraham de  
Moivre



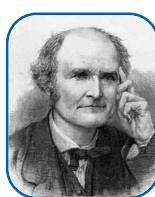
Leonhard Euler



Gottfried  
Wilhelm  
Leibniz



Herbert Wilf



Arthur Cayley



Augustin-Louis  
Cauchy

# What is Combinatorics?

**Definition 1:** *Combinatorics* is the branch of mathematics that studies the existence, enumeration, and structure of *finite* discrete objects.

Three fundamental questions arise in every combinatorial problem:

## Existence

Does an arrangement of the required kind *exist* at all?

*Example:* Does a graph with 5 vertices where every vertex has exactly 3 neighbors exist?

## Enumeration

*How many* such arrangements are there?

*Example:* How many 4-digit PIN codes are possible?

## Optimization

Which arrangement is *best* by some criterion?

*Example:* What is the shortest tour visiting all  $n$  cities exactly once?

# Basic Counting Principles

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*“Music is the pleasure the human mind experiences from counting without being aware that it is counting.”*

*— Gottfried Wilhelm Leibniz*

# Basic Counting Rules

## Addition (*either/or*)

If  $S$  splits into *disjoint* parts  $S_1, \dots, S_k$ , count each part separately:  $|S| = |S_1| + \dots + |S_k|$

*Example:* 26 vowels + 20 consonants = 46 letters.

## Subtraction (complement)

Count what you *don't* want and subtract from the universe  $U$ :  $|\overline{S}| = |U| - |S|$

*Example:* 8-bit strings *with* a leading zero:  $2^8 - 2^7 = 128$ .

## Multiplication (*and*)

For *independent* sequential choices of sizes  $n_1, \dots, n_k$ , multiply:  $|S| = n_1 \cdot n_2 \cdot \dots \cdot n_k$

*Example:*  $26 \times 10 = 260$  two-character codes (letter then digit).

## Bijection (one-to-one)

If  $S$  and  $T$  are in *bijective correspondence*, they have the same size:  $|S| = |T|$

*Example:*  $\binom{n}{k} = \binom{n}{n-k}$  via  $A \mapsto [n] \setminus A$ .

## Addition Principle

**Definition 2:** We say a finite set  $S$  is *partitioned* into *parts*  $S_1, \dots, S_k$  if the parts are pairwise disjoint and their union is  $S$ . In other words,  $S_i \cap S_j = \emptyset$  for  $i \neq j$  and  $S_1 \cup S_2 \cup \dots \cup S_k = S$ . In that case:

$$|S| = |S_1| + |S_2| + \dots + |S_k|$$

*Example:* Let  $S$  be the set of students attending the combinatorics lecture. It can be partitioned into parts  $S_1$  and  $S_2$  where

$S_1$  = set of students that *like* easy examples.

$S_2$  = set of students that *don't like* easy examples.

If  $|S_1| = 22$  and  $|S_2| = 8$ , then we can conclude  $|S| = |S_1| + |S_2| = 30$ .

## Multiplication Principle

**Definition 3:** If  $S$  is a finite set that is the *product* of  $S_1, \dots, S_k$ , that is,  $S = S_1 \times \dots \times S_k$ , then

$$|S| = |S_1| \times \dots \times |S_k|$$

*Example:* A *license plate* consists of 3 letters followed by 3 digits. The letters are chosen from  $\{A, \dots, Z\}$  (26 choices each) and the digits from  $\{0, \dots, 9\}$  (10 choices each). Since each position is filled independently:

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 = 26^3 \cdot 10^3 = 17,576,000$$

## Subtraction Principle

**Definition 4:** Let  $S$  be a subset of a finite set  $T$ . We define the *complement* of  $S$  as  $\overline{S} = T \setminus S$ . Then

$$|\overline{S}| = |T| - |S|$$

*Example:* Let  $T$  be the set of all students at a university ( $|T| = 23905$ ). Let  $S \subseteq T$  be the set of students studying *neither* mathematics nor computer science ( $|S| = 20178$ ). The complement  $\overline{S} = T \setminus S$  consists of students studying mathematics or computer science:

$$|\overline{S}| = |T| - |S| = 23905 - 20178 = 3727$$

## Bijection Principle

**Definition 5:** If  $S$  and  $T$  are sets, then

$$|S| = |T| \iff \text{there exists a bijection between } S \text{ and } T$$

*Example:* The number of  $k$ -subsets of  $[n]$  equals the number of  $(n - k)$ -subsets:  $\binom{n}{k} = \binom{n}{n-k}$ . The map  $S \mapsto [n] \setminus S$  is a bijection from  $\binom{[n]}{k}$  to  $\binom{[n]}{n-k}$ : each  $k$ -subset maps to its  $(n - k)$ -element complement, and the map is its own inverse.

**Note:** The bijection principle works for *infinite* sets too — this is the foundation of Cantor's theory of cardinality.

## Pigeonhole Principle

**Definition 6:** Let  $S_1, \dots, S_k$  be finite sets that are pairwise disjoint and  $|S_1| + |S_2| + \dots + |S_k| = n$ .

$$\exists i \in \{1, \dots, k\} : |S_i| \geq \left\lceil \frac{n}{k} \right\rceil \quad \text{and} \quad \exists j \in \{1, \dots, k\} : |S_j| \leq \left\lfloor \frac{n}{k} \right\rfloor$$

*Example:* Assume there are 5 holes in the wall where pigeons nest. Say there is a set  $S_i$  of pigeons nesting in hole  $i$ . Assume there are  $n = 17$  pigeons in total. Then we know:

- There is some hole with at least  $d = 4$  pigeons.
- There is some hole with at most  $b = 3$  pigeons.

**Birthday attack:** a hash function maps arbitrary inputs to  $n$ -bit digests.

- With  $2^n + 1$  inputs, at least two must collide — guaranteed by the pigeonhole principle.
- With only  $\approx 2^{n/2}$  random inputs, collision probability already exceeds 50%.

This is the *birthday bound* and sets the security level of hash functions in cryptography.

## Double Counting

If we count the same quantity in *two different ways*, then this gives us a (perhaps non-trivial) identity.

*Example (Handshaking Lemma):* Assume there are  $n$  people at a party and everybody will shake hands with everybody else. How many handshakes will occur? We count this number in two ways:

1. Every person shakes  $n - 1$  hands and there are  $n$  people. However, two people are involved in a handshake so if we just multiply  $n \cdot (n - 1)$ , then every handshake is counted twice. The total number of handshakes is therefore  $\frac{n \cdot (n-1)}{2}$ .
2. We number the people from 1 to  $n$ . To avoid counting a handshake twice, we count for person  $i$  only the handshakes with persons of lower numbers. Then the total number of handshakes is:

$$\sum_{i=1}^n (i - 1) = \sum_{i=0}^{n-1} i = \sum_{i=1}^{n-1} i$$

The identity we obtain is therefore:  $\sum_{i=1}^{n-1} i = \frac{n \cdot (n - 1)}{2}$

# Arrangements, Permutations, Combinations

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*“As for everything else, so for a mathematical theory:  
beauty can be perceived but not explained.”*

*— Arthur Cayley*

# Ordered Arrangements

**Definition 7:** Denote by  $[n] = \{1, \dots, n\}$  the set of natural numbers from 1 to  $n$ .

Hereinafter, let  $X$  be a finite set.

**Definition 8:** An *ordered arrangement* of  $n$  elements of  $X$  is a *map*  $s : [n] \rightarrow X$ .

- Here,  $[n]$  is the *domain* of  $s$ , and  $s(i)$  is the *image* of  $i \in [n]$  under  $s$ .
- The set  $\{x \in X \mid s(i) = x \text{ for some } i \in [n]\}$  is the *range* of  $s$ .

Other common names for ordered arrangements are:

- *string* (or *word*), e.g. “Banana”
- *sequence*, e.g. “0815422372”
- *tuple*, e.g.  $(3, 5, 2, 5, 8)$

Example:

| $i$    | 1                                                                                 | 2                                                                                 | 3                                                                                 | 4                                                                                 | 5                                                                                 | 6                                                                                 | 7                                                                                 |
|--------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| $s(i)$ |  |  |  |  |  |  |  |

# Permutations

**Definition 9:** A *permutation* of  $X$  is a *bijective* map  $\pi : [n] \rightarrow X$ .

Usually,  $X = [n]$ , and the set of all permutations of  $[n]$  is denoted by  $S_n$ .

Example:

|          |   |   |   |   |   |   |   |
|----------|---|---|---|---|---|---|---|
| $i$      | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| $\pi(i)$ | 2 | 7 | 1 | 3 | 5 | 4 | 6 |

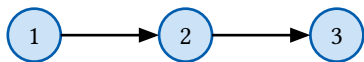
**Definition 10:** *k-permutation* of  $X$  is an ordered arrangement of  $k$  *distinct* elements of  $X$ , that is, an *injective* map  $\pi : [k] \rightarrow X$ .

The set of all  $k$ -permutations of  $X = [n]$  is denoted by  $P(n, k)$ . In particular,  $S_n = P(n, n)$ .

## Circular Permutations

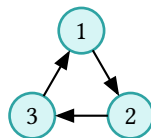
**Definition 11:** A *circular  $k$ -permutation* of  $[n]$  arranges  $k$  distinct elements of  $[n]$  in a circle. Two such arrangements are considered the same if one is a cyclic rotation of the other. The set of all circular  $k$ -permutations of  $[n]$  is denoted  $P_{c(n,k)}$ .

**Linear:** all  $3! = 6$  are distinct:



$(1, 2, 3), (1, 3, 2), (2, 1, 3),$   
 $(2, 3, 1), (3, 1, 2), (3, 2, 1)$

**Circular:** rotations are identical:



$(1, 2, 3) \sim (2, 3, 1) \sim (3, 1, 2)$  – same circle

Only 2 distinct circles:  $(1, 2, 3)$  and  $(1, 3, 2)$

## Counting Permutations

**Theorem 1:** For any natural numbers  $0 \leq k \leq n$ , we have

$$|P(n, k)| = n \cdot (n - 1) \cdot \dots \cdot (n - k + 1) = \frac{n!}{(n - k)!}$$

This formula is also called the *falling factorial* and denoted  $n^{\underline{k}}$  or  $(n)_k$ .

**Proof:** A permutation is an injective map  $\pi : [k] \rightarrow [n]$ . We count the number of ways to pick such a map, picking the images one after the other. There are  $n$  ways to choose  $\pi(1)$ . Given a value for  $\pi(1)$ , there are  $(n - 1)$  ways to choose  $\pi(2)$  (since we may not choose  $\pi(1)$  again). Continuing like this, there are  $(n - i + 1)$  ways to pick  $\pi(i)$ , and the last value we pick is  $\pi(k)$  with  $(n - k + 1)$  possibilities.

Every  $k$ -permutation can be constructed like this in *exactly one way*. The total number of  $k$ -permutations is therefore given as the product:

$$|P(n, k)| = n \cdot (n - 1) \cdot \dots \cdot (n - k + 1) = \frac{n!}{(n - k)!}$$

□

## Counting Circular Permutations

**Theorem 2:** For any natural numbers  $0 \leq k \leq n$ , we have

$$|P_c(n, k)| = \frac{n!}{k \cdot (n - k)!}$$

**Proof:** We doubly count  $P(n, k)$ :

1.  $|P(n, k)| = \frac{n!}{(n-k)!}$  which we proved before.
2.  $|P(n, k)| = |P_c(n, k)| \cdot k$  because every equivalence class in  $P_c(n, k)$  contains  $k$  permutations from  $P(n, k)$  since there are  $k$  ways to rotate a  $k$ -permutation.

From this we get  $\frac{n!}{(n-k)!} = |P_c(n, k)| \cdot k$ , which implies  $|P_c(n, k)| = \frac{n!}{k \cdot (n-k)!}$ . □

# Unordered Arrangements

**Definition 12:** An *unordered arrangement* of  $k$  elements of  $X$  is a *multiset*  $S = \langle X, r \rangle$  of size  $k$ .

In a multiset,  $X$  is the set of *types*, and for each type  $x \in X$ ,  $r_x$  is its *repetition number*.

*Example:* Let  $X = \{ \text{👉}, \text{👈}, \text{🐱}, \text{🎴}, \text{🌵} \}$ .

- An unordered arrangement of 7 elements could be  $S = \{ \text{👉}, \text{👉}, \text{👈}, \text{🐱}, \text{🐱}, \text{🐱}, \text{🌵} \}^*$ .
- The same multiset could be written as  $S = \{ 2 \text{👉}, 1 \text{👈}, 3 \text{🐱}, 0 \text{🎴}, 1 \text{🌵} \}$ .

## Subsets

The most important special case of unordered arrangements is where all repetitions are 1, that is,  $r_x = 1$  for all  $x \in X$ . Then  $S$  is simply a *subset* of  $X$ , denoted  $S \subseteq X$ .

**Definition 13:** A *k-combination* of  $X$  is an unordered arrangement of  $k$  *distinct* elements of  $X$ .

**Note:** The more standard term is *subset*. The term “combination” is only used to emphasize the selection process.

The set of all  $k$ -subsets of  $X$  is denoted  $\binom{X}{k} = \{A \subseteq X \mid |A| = k\}$ . If  $|X| = n$ , then

$$\binom{n}{k} := \left| \binom{X}{k} \right|$$

*Example:* The set of edges in a simple undirected graph consists of 2-subsets of its vertices:  $E \subseteq \binom{V}{2}$ .

## Counting $k$ -Combinations

**Theorem 3:** For  $0 \leq k \leq n$ , we have

$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$$

**Proof:**  $|P(n, k)| = \frac{n!}{(n-k)!} = \binom{n}{k} \cdot k!$

□

## Four Counting Problems

|                                        | No repetition                               | With repetition    |
|----------------------------------------|---------------------------------------------|--------------------|
| <b>Ordered</b> ( $k$ -perm., sequence) | $\frac{n!}{(n-k)!} = n^{\underline{k}}$     | $n^k$              |
| <b>Unordered</b> ( $k$ -comb., subset) | $\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!}$ | $\binom{k+n-1}{k}$ |

- $n$  is the size of the ground set  $X$ .
- $k$  is the size of the arrangement we want to count.
- “Ordered” means position matters:  $(A, B) \neq (B, A)$ .
- “Unordered” means only the selection matters:  $\{A, B\} = \{B, A\}$ .
- “With repetition” means the same element can appear multiple times.

# **Multisets**

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# Multiset

**Definition 14:** A *multiset* is a modification of the concept of a set that allows for *repetitions* of its elements. Formally, it is denoted as a pair  $M = \langle X, r \rangle$ , where  $X$  is the *groundset* (the set of *types*) and  $r : X \rightarrow \mathbb{N}_0$  is the *multiplicity function*.

*Example:* When the multiset is defined by enumeration, it is advisable to use the notation with the star:

$$M = \{a, b, a, a, b\}^* = \{3 \cdot a, 2 \cdot b\} \quad X = \{a, b\} \quad r_a = 3, r_b = 2$$

*Example:* Prime factorization of a natural number  $n$  is a multiset, e.g.  $120 = 2^3 \cdot 3^1 \cdot 5^1$ .

## $k$ -Combinations of a Multiset

**Definition 15:** Let  $X$  be a finite set of types, and let  $M = \langle X, r \rangle$  be a finite multiset with repetition numbers  $r_1, \dots, r_{|X|}$ . A  *$k$ -combination of  $M$*  is a multiset  $S = \langle X, s \rangle$  with types in  $X$  and repetition numbers  $s_1, \dots, s_{|X|}$  such that  $s_i \leq r_i$  for all  $1 \leq i \leq |X|$ , and  $\sum_{i=1}^{|X|} s_i = k$ .

*Example:* Consider  $M = \{2 \text{ 🦊}, 1 \text{ 🥤}, 3 \text{ 🍉}, 1 \text{ 💎}\}$ .

- $T = \{1 \text{ 🦊}, 2 \text{ 🍉}\}$  is a 3-combination of  $M$ .
- $T' = \{3 \text{ 💎}\}$  is not.

*Counting  $k$ -combinations of a multiset is not as simple as it might seem...*

## $k$ -Permutations of a Multiset

**Definition 16:** Let  $M$  be a finite multiset with set of types  $X$ . A  *$k$ -permutation of  $M$*  is an ordered arrangement of  $k$  elements of  $M$  where different orderings of elements of the same type are *not distinguished*. This is an ordered multiset with types in  $X$  and repetition numbers  $s_1, \dots, s_{|X|}$  such that  $s_i \leq r_i$  for all  $1 \leq i \leq |X|$ , and  $\sum_{i=1}^{|X|} s_i = k$ .

**Note:** There might be several elements of the same type compared to a permutation of a set (where each repetition number equals 1).

*Example:* Let  $M = \{2 \text{ 🦊}, 1 \text{ 🥤}, 3 \text{ 🍉}, 1 \text{ 💎}\}$ , then  $T = (\text{💎}, \text{🍉}, \text{🍉}, \text{🦊})$  is a 4-permutation of multiset  $M$ .

## Binomial Theorem

**Theorem 4:** The expansion of any non-negative integer power  $n$  of the binomial  $(x + y)$  is a sum

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} \cdot x^k \cdot y^{n-k}$$

where each  $\binom{n}{k}$  is a positive integer known as a *binomial coefficient*, defined as

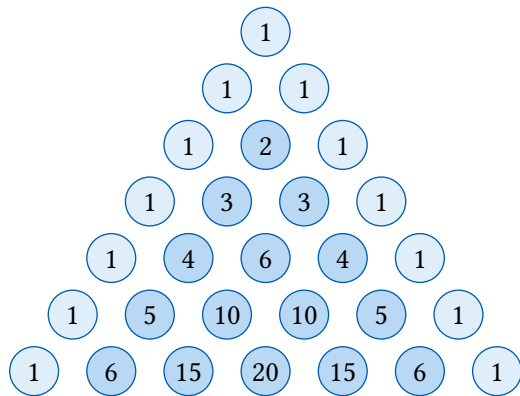
$$\binom{n}{k} = \frac{n!}{k! \cdot (n-k)!} = \frac{n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1)}{k \cdot (k-1) \cdot (k-2) \cdot \dots \cdot 2 \cdot 1}$$

# Pascal's Triangle

Row  $n$  of *Pascal's triangle* lists  $\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n}$ . Each entry is the sum of the two entries above it:  $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ .

**Key identities:**

- **Row sum:**  $\sum_{k=0}^n \binom{n}{k} = 2^n$  (set  $x = y = 1$ )
- **Symmetry:**  $\binom{n}{k} = \binom{n}{n-k}$
- **Absorption:**  $n \binom{n-1}{k-1} = (k+1) \binom{n}{k+1}$
- **Vandermonde:**  $\sum_k \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$



# Multinomial Theorem

**Theorem 5:** The generalization of the binomial theorem:

$$(x_1 + \dots + x_r)^n = \sum_{\substack{0 \leq k_1, \dots, k_r \leq n \\ k_1 + \dots + k_r = n}}^n \binom{n}{k_1, \dots, k_r} \cdot x_1^{k_1} \cdot \dots \cdot x_r^{k_r}$$

*Multinomial coefficients* are defined as

$$\binom{n}{k_1, \dots, k_r} = \frac{n!}{k_1! \cdot \dots \cdot k_r!}$$

**Note:** Binomial coefficients are special cases of multinomial coefficients ( $r = 2$ ):

$$\binom{n}{k} = \binom{n}{k_1, k_2} = \binom{n}{k, n-k} = \frac{n!}{k! \cdot (n-k)!}$$

## Multinomial Theorem [2]

**Proof:** Expand  $(x_1 + \dots + x_r)^n$  by choosing one term from each of the  $n$  factors. A monomial  $x_1^{k_1} \cdot \dots \cdot x_r^{k_r}$  with  $k_1 + \dots + k_r = n$  is produced whenever we choose  $x_i$  from exactly  $k_i$  of the  $n$  factors. The number of ways to assign the  $n$  factor-positions to the  $r$  variables in groups of sizes  $k_1, \dots, k_r$  is

$$\binom{n}{k_1, \dots, k_r} = \frac{n!}{k_1! \cdot \dots \cdot k_r!}.$$

□

**Theorem 6:** Let  $S$  be a finite multiset with  $k$  different types and repetition numbers  $r_1, \dots, r_k$ . Let the size of  $S$  be  $n = r_1 + \dots + r_k$ . Then the number of  $n$ -permutations of  $S$  equals

$$\binom{n}{r_1, \dots, r_k}$$

**Proof:** In an  $n$ -permutation there are  $n$  positions that need to be assigned a type.

First, choose the  $r_1$  positions for the first type, there are  $\binom{n}{r_1}$  ways to do so. Then, assign  $r_2$  positions for the second type, out of the  $(n - r_1)$  positions that are still available, there are  $\binom{n-r_1}{r_2}$  ways to do so. Continue for all  $k$  types. The total number of choices will be:

## Multinomial Theorem [3]

$$\binom{n}{r_1} \cdot \binom{n-r_1}{r_2} \cdot \dots \cdot \binom{n-r_1-r_2-\dots-r_{k-1}}{r_k} = \binom{n}{r_1, \dots, r_k}$$

□

## $k$ -Combinations of an *Infinite Multiset*

*Example:* Suppose you have a *sufficiently large* amount of each type of fruit (🍌, 🍎, 🍐) in the supermarket, and you want to buy *two* fruits. How many choices do you have?

There are exactly *six* combinations: {🍌, 🍌}, {🍌, 🍎}, {🍌, 🍐}, {🍎, 🍎}, {🍎, 🍐}, {🍐, 🍐}.

Note that your selection is *not ordered*, so {🍐, 🍎} and {🍎, 🍐} are considered the *same* choice.

## $k$ -Combinations of an *Infinite Multiset* [2]

**Theorem 7:** Let  $k, s \in \mathbb{N}$  and let  $S$  be a multiset with  $s$  types and large repetition numbers (each  $r_1, \dots, r_s$  is *at least  $k$* ), then the number of  $k$ -combinations of  $S$  equals

$$\binom{k+s-1}{k} = \binom{k+s-1}{s-1}$$

**Proof:** Let  $S = \langle X, r_\infty \rangle = \{\infty \text{ 🍌}, \infty \text{ 🍎}, \infty \text{ 🍇}\}$  with  $r_x = \infty$  and  $|X| = s = 3$ .

- Let  $k = 5$  (as an example). Consider a 5-combination of  $S$ : { 🍌 🍌 🍌 🍌 🍌, 🍎, 🍌, 🍇, 🍇 }.
- Reorder and group: { 🍌 🍌 🍌 | 🍎 | 🍇 🍇 }.
- Convert to *dots* and *bars*:    ••• | • | ••
- Represent as a 2-type multiset:  $M = \{k \cdot \bullet, (s-1) \cdot |\}$
- Observe: each *permutation* of  $k$  dots and  $(s-1)$  bars corresponds *uniquely* to a  *$k$ -combination* of  $S$ .
- Permute the 2-type multiset:  $\binom{k+s-1}{k, s-1}$  ways, by Theorem 5.

This method is also known as *Stars and Bars*. □

# **Compositions**

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## Weak Compositions

**Definition 17:** A *weak composition* of a non-negative integer  $k \geq 0$  into  $s$  parts is a *solution* to the equation  $b_1 + \dots + b_s = k$ , where each  $b_i \geq 0$ .

*Example:* Let  $k = 5$ ,  $s = 3$ . Possible non-negative integer solutions for  $b_1 + b_2 + b_3 = 5$  are:

- $(b_1, b_2, b_3) = (1, 1, 3)$
- $(b_1, b_2, b_3) = (1, 3, 1)$
- $(b_1, b_2, b_3) = (2, 0, 3)$
- $(b_1, b_2, b_3) = (0, 5, 0)$
- ... (total 21 solutions)

**Note:** If  $M$  is a multiset over groundset  $\{1, \dots, s\}$  with all multiplicities infinite ( $r_i = \infty$ ), then for  $k \geq 0$ , the number of sub-multisets of  $M$  of size  $k$  is exactly the number of weak compositions of  $k$  into  $s$  parts.

## Counting Weak Compositions

**Theorem 8:** There are  $\binom{k+s-1}{k, s-1}$  *weak compositions* of  $k > 0$  into  $s$  parts.

**Proof:** Observe that  $k = \overbrace{1+1+\dots+1+1}^{k \text{ ones}}$   
 $\qquad\qquad\qquad \underbrace{\qquad}_{b_1} \quad \underbrace{\qquad}_{b_i} \quad \underbrace{\qquad}_{b_s}$

Use the *stars-and-bars* method to count the number of  $s$  groups composed of  $k$  “ones”. □

*Example:* Let  $k = 3$ . There are  $\binom{3+3-1}{3, 3-1} = \binom{5}{3} = \binom{5}{2} = 10$  ways to decompose  $k = 3$  into  $s = 3$  parts:

$$\begin{aligned} k = 3 &= \\ &= 0 + 1 + 2 = 0 + 2 + 1 \\ &= 1 + 0 + 2 = 1 + 2 + 0 = 1 + 1 + 1 \\ &= 2 + 0 + 1 = 2 + 1 + 0 \\ &= 3 + 0 + 0 = 0 + 3 + 0 = 0 + 0 + 3 \end{aligned}$$

## Compositions

**Definition 18:** A *composition* of a positive integer  $k \geq 1$  into  $s$  *positive* parts is a *solution* to the equation  $b_1 + \dots + b_s = k$ , where each  $b_i > 0$ .

**Theorem 9:** There are  $\binom{k-1}{s-1}$  *compositions* of  $k > 0$  into  $s$  positive parts.

**Proof:** Line up  $k$  ones in a row. A composition  $b_1 + \dots + b_s = k$  with each  $b_i \geq 1$  corresponds to choosing  $s - 1$  of the  $k - 1$  *gaps between consecutive ones* as dividers. There are  $\binom{k-1}{s-1}$  such choices.  $\square$

## Compositions [2]

**Theorem 10:** The total number of compositions of  $k > 0$  into *some* number of positive parts is

$$\sum_{s=1}^k \binom{k-1}{s-1} = 2^{k-1}$$

**Proof:** Substituting  $j = s - 1$ :

$$\sum_{s=1}^k \binom{k-1}{s-1} = \sum_{j=0}^{k-1} \binom{k-1}{j} = (1+1)^{k-1} = 2^{k-1}$$

by the Binomial Theorem applied to  $(1+1)^{k-1}$ . □

## Parallel Summation Identity

Q: How many integer solutions are there to the *inequality*  $b_1 + \dots + b_s \leq k$ , where each  $b_i \geq 0$ ?

**Theorem 11:** 
$$\sum_{m=0}^k \binom{m+s-1}{m} = \binom{k+s}{k}$$

**Proof** (*hint*): Introduce a “dummy” variable  $b_{s+1}$  to take up the *slack* between  $b_1 + \dots + b_s$  and  $k$ . Construct a bijection with the solutions to  $b_1 + \dots + b_s + b_{s+1} = k$ , where  $b_i \geq 0$ . □

# **Set Partitions**

---

## Set Partitions

**Definition 19:** A *partition* of a set  $X$  is a set of non-empty subsets of  $X$  such that every element of  $X$  belongs to exactly one of these subsets.

Equivalently, a family of sets  $P$  is a partition of  $X$  iff:

1. The family  $P$  does not contain the empty set:  $\emptyset \notin P$ .
2. The union of  $P$  is  $X$ , that is,  $\bigcup_{A \in P} A = X$ . The sets in  $P$  are said to *cover*  $X$ .
3. The intersection of any two distinct sets in  $P$  is empty:  $\forall A, B \in P. (A \neq B) \rightarrow (A \cap B = \emptyset)$ .  
The sets in  $P$  are said to be *pairwise disjoint* or *mutually exclusive*.

The sets in  $P$  are called *blocks*, *parts*, or *cells*, of the partition.

The block in  $P$  containing an element  $x \in X$  is denoted by  $[x]$ .

## Examples of Set Partitions

*Example:* The empty set  $X = \emptyset$  has exactly one partition,  $P = \emptyset$ .

*Example:* Any singleton set  $X = \{x\}$  has exactly one partition,  $P = \{\{x\}\}$ .

*Example:* For any non-empty proper subset  $A \subset U$ , the set  $A$  and its complement form a partition of  $U$ , namely  $P = \{A, U - A\}$ .

*Example:* The set  $X = \{1, 2, 3\}$  has five partitions:

1.  $\{\{1\}, \{2\}, \{3\}\}$  or  $1 \mid 2 \mid 3$
2.  $\{\{1\}, \{2, 3\}\}$  or  $1 \mid 2 \ 3$
3.  $\{\{1, 2\}, \{3\}\}$  or  $1 \ 2 \mid 3$
4.  $\{\{1, 3\}, \{2\}\}$  or  $1 \ 3 \mid 2$
5.  $\{\{1, 2, 3\}\}$  or  $1 \ 2 \ 3$

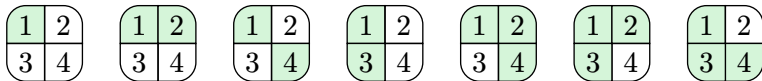
*Example:* The following are *not* partitions of  $\{1, 2, 3\}$ :

- $\{\{\}, \{1, 3\}, \{2\}\}$ , because it contains the empty set.
- $\{\{1, 2\}, \{2, 3\}\}$ , because the element 2 is contained in more than one block.
- $\{\{1\}, \{3\}\}$ , because no block contains the element 3.

## Counting Set Partitions

**Definition 20:** The number of partitions of a set  $X$  (of size  $n = |X|$ ) into  $k$  non-empty blocks (“unlabeled subsets”) is called a *Stirling number of the second kind* and denoted  $S(n, k)$  or  $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ .

*Example:* Let  $X = \{1, 2, 3, 4\}$ ,  $k = 2$ . There are 7 possible partitions:



## Counting Set Partitions [2]

**Theorem 12:** Let  $\left\{ \begin{smallmatrix} n \\ 0 \end{smallmatrix} \right\} = 0$  for  $n \geq 1$ ,  $\left\{ \begin{smallmatrix} 0 \\ k \end{smallmatrix} \right\} = 0$  for  $k \geq 1$ , and  $\left\{ \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right\} = 1$ . For  $n, k \geq 1$ , we have:

$$\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\} + k \cdot \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\}$$

**Proof (informal):** Consider the element  $n$  in any partition of  $[n]$  into  $k$  blocks.

There are exactly two cases:

1.  $\{n\}$  forms a singleton block on its own.

The remaining  $n - 1$  elements are partitioned into  $k - 1$  non-empty blocks:  $\left\{ \begin{smallmatrix} n-1 \\ k-1 \end{smallmatrix} \right\}$  ways.

2.  $n$  joins an existing block of a partition of  $[n - 1]$ .

Start with any partition of  $[n - 1]$  into  $k$  blocks ( $\left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\}$  ways),

then insert  $n$  into one of the  $k$  blocks:  $k \cdot \left\{ \begin{smallmatrix} n-1 \\ k \end{smallmatrix} \right\}$  ways.

These two cases are exhaustive and disjoint.



## Bell Numbers

**Definition 21:** The total number of partitions of a set  $X$  of size  $n = |X|$  (into an arbitrary number of non-empty blocks) is called a *Bell number* and denoted  $B_n$ .

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

**Note:** Consider the special case of  $n = 0$ . There is exactly *one* partition of  $\emptyset$  into non-empty parts:  $\emptyset = \bigcup_{A \in \emptyset} A \in \emptyset$ . Every  $A \in \emptyset$  is non-empty, since no such  $A$  exists. Thus, we have  $B_0 = S(0, 0) = 1$ .

## Bell Numbers [2]

**Theorem 13:** For  $n \geq 1$ , we have the recursive identity for Bell numbers:

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k$$

**Proof:** Every partition of  $[n]$  has one part that contains the number  $n$ . In addition to  $n$ , this part also contains  $k$  other numbers (for some  $0 \leq k \leq n-1$ ). The remaining  $n-1-k$  elements are partitioned arbitrarily. From this correspondence, we obtain the desired identity:

$$B_n = \sum_{k=0}^{n-1} \binom{n-1}{k} B_{n-1-k} = \sum_{k=0}^{n-1} \binom{n-1}{n-1-k} B_{n-1-k} = \sum_{k=0}^{n-1} \binom{n-1}{k} B_k$$

□

# **Integer Partitions**

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## Integer Partitions

The integer 4 can be written as a sum of positive integers in exactly 5 ways:  $4 = 3 + 1 = 2 + 2 = 2 + 1 + 1 = 1 + 1 + 1 + 1$ . Order does *not* matter:  $3 + 1$  and  $1 + 3$  are the same partition.

**Definition 22:** An *integer partition* of a positive integer  $n \geq 1$  into  $k$  *positive* parts is a solution to the equation  $n = a_1 + \dots + a_k$ , where  $a_1 \geq a_2 \geq \dots \geq a_k \geq 1$ .

The number of partitions into exactly  $k$  parts is  $p_k(n)$ , defined recursively:

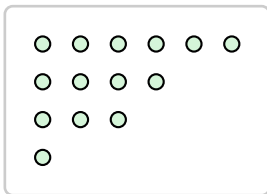
$$p_k(n) = \begin{cases} 0 & \text{if } k > n \\ 0 & \text{if } n \geq 1 \text{ and } k = 0 \\ 1 & \text{if } n = k = 0 \\ p_k(n - k) + p_{k-1}(n - 1) & \text{if } 1 \leq k \leq n \end{cases}$$

The *partition function*  $p(n) = \sum_{k=0}^n p_k(n)$  counts all partitions of  $n$ .

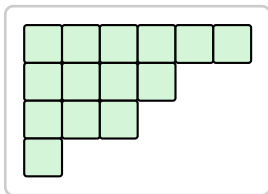
# Ferrers Diagrams and Young Tableaux

*Example:* Consider an integer partition:  $14 = 6 + 4 + 3 + 1$ .

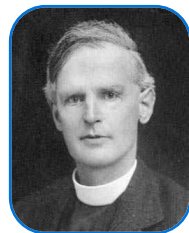
**Ferrer Diagram**



**Young Tableaux**



Norman Ferrer



Alfred Young

# Inclusion–Exclusion

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*“There are very few things which we know, which are not capable of being reduced to a mathematical reasoning.”*

*— Abraham de Moivre, The Doctrine of Chances (1718)*

## Motivating Example

*Example:* How many integers in  $\{1, \dots, 100\}$  are divisible by 2 *or* by 3?

- Let  $A = \{k \in [100] \mid 2 \mid k\}$ , so  $|A| = 50$ .
- Let  $B = \{k \in [100] \mid 3 \mid k\}$ , so  $|B| = 33$ .
- Their intersection  $A \cap B = \{k \in [100] \mid 6 \mid k\}$  has  $|A \cap B| = 16$ .

By inclusion-exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B| = 50 + 33 - 16 = 67$$

Generalizing this to an arbitrary number of sets gives the full PIE theorem.

## Principle of Inclusion–Exclusion (PIE)

**Theorem 14:** Let  $X$  be a finite set and  $P_1, \dots, P_m$  *properties*.

- Define  $X_i = \{x \in X \mid x \text{ has } P_i\}$ , i.e. the set of all elements from  $X$  having a property  $P_i$ .
- Define for  $S \subseteq [m]$  the set  $N(S) = \{x \in X \mid \forall i \in S : x \text{ has } P_i\}$ . Observe:  $N(S) = \bigcap_{i \in S} X_i$ .

The number of elements of  $X$  that satisfy *none* of the properties  $P_1, \dots, P_m$  is given by

$$|X \setminus (X_1 \cup \dots \cup X_m)| = \sum_{S \subseteq [m]} (-1)^{|S|} |N(S)| \quad (1)$$

**Proof:** Consider any  $x \in X$ . If  $x \in X$  has none of the properties, then  $x \in N(\emptyset)$  and  $x \notin N(S)$  for any other  $S \neq \emptyset$ . Hence  $x$  *contributes 1* to the sum (1).

If  $x \in X$  has exactly  $k \geq 1$  of the properties, call this set  $T \in \binom{[m]}{k}$ . Then  $x \in N(S)$  iff  $S \subseteq T$ .

The *contribution of  $x$*  to the sum (1) is  $\sum_{S \subseteq T} (-1)^{|S|} = \sum_{i=0}^k \binom{k}{i} (-1)^i = 0$ , i.e. *zero*. □

**Note:** In the last step, we used that for any  $y \in \mathbb{R}$  we have  $(1 - y)^k = \sum_{i=0}^k \binom{k}{i} (-y)^i$  which implies (for  $y = 1$ ) that  $0 = \sum_{i=0}^k \binom{k}{i} (-1)^i$ .

## Counting the Union

**Corollary 14.1:**

$$\left| \bigcup_{i \in [m]} X_i \right| = \sum_{\emptyset \neq S \subseteq [m]} (-1)^{|S|-1} |N(S)|$$

# Applications of PIE

## Step 1: Define “bad” properties

Identify elements of a universe  $X$  to *exclude*: those having *at least one* of properties  $P_1, \dots, P_m$ .

Target: count  $X \setminus (X_1 \cup \dots \cup X_m)$ .

## Step 2: Count $N(S)$

For each  $S \subseteq [m]$ , determine  $N(S)$  – number of elements of  $X$  having *all* bad properties  $P_i$  for  $i \in S$ .

## Step 3: Apply PIE

Use Theorem 14 to compute a closed formula:

$$|X \setminus (X_1 \cup \dots \cup X_m)| = \sum_{S \subseteq [m]} (-1)^{|S|} N(S)$$

## Counting Surjections via PIE

**Theorem 15:** The number of surjections from  $[k]$  to  $[n]$  is given by

$$\left| \left\{ f : [k] \xrightarrow{\text{surj.}} [n] \right\} \right| = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^k$$

**Proof:** Let  $X$  be the set of all maps from  $[k]$  to  $[n]$ .

1. *Define bad properties:* Define the “bad” property  $P_i$  for  $i \in [n]$  as “ $i$  is not in the image of  $f$ ”, i.e.

$$f : [k] \rightarrow [n] \text{ has property } P_i \iff \forall j \in [k] : f(j) \neq i$$

The *surjective* functions are exactly those functions that *do not* have any of the “bad” properties.

2. *Count  $N(S)$ :* We claim  $N(S) = (n - |S|)^k$  for any  $S \subseteq [n]$ . To see this, observe that  $f$  has all properties with indices from  $S$  if and only if  $f(i) \notin S$  for all  $i \in [k]$ . In other words,  $f$  must be a function from  $[k]$  to  $[n] \setminus S$ , and there are  $(n - |S|)^k$  of those.

## Counting Surjections via PIE [2]

3. *Apply PIE:* Using Theorem 14, the number of surjections from  $[k]$  to  $[n]$  is

$$\begin{aligned} |X \setminus (X_1 \cup \dots \cup X_n)| &\stackrel{\text{PIE}}{=} \sum_{S \subseteq [n]} (-1)^{|S|} |N(S)| \\ &= \sum_{S \subseteq [n]} (-1)^{|S|} (n - |S|)^k \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i} (n - i)^k \end{aligned}$$

In the last step, we used that  $(-1)^{|S|} (n - |S|)^k$  only depends on the size of  $S$ , and there are  $\binom{n}{i}$  sets  $S \subseteq [n]$  of size  $i$ .

□

## Useful Corollaries

**Corollary 15.1:** Consider the case  $n = k$ . A function from  $[n]$  to  $[n]$  is a *surjection* iff it is a *bijection*. Since there are  $n!$  bijections on  $[n]$  (namely, all permutations), we have the following identity:

$$n! = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^n$$

**Corollary 15.2:** A surjection from  $[k]$  to  $[n]$  can be seen as a partition of  $[k]$  into  $n$  non-empty distinguishable (labeled) parts (the map assigns a part to each  $i \in [n]$ ).

Since the partition of  $[k]$  into  $n$  non-empty indistinguishable parts is denoted  $s_n^{\text{II}}(k)$ , and there are  $n!$  ways to assign labels to  $n$  parts, we obtain that the number of surjections is equal to  $n!s_n^{\text{II}}(k)$ :

$$n!s_n^{\text{II}}(k) = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)^k$$

## Derangements

**Theorem 16:** The *derangements*  $D_n$  on  $n$  elements are permutations of  $[n]$  without fixed points.

The number of derangements is given by

$$|D_n| = \sum_{i=0}^n (-1)^i \binom{n}{i} (n-i)!$$

**Proof:** Let  $X$  be the set of all permutations of  $[n]$ .

1. Define the “bad” property  $P_i$  to mean “ $\pi$  has a fixpoint  $i$ ” ( $i \in [n]$ ):

$$\pi \in X \text{ has property } P_i \iff \pi(i) = i$$

2. We claim  $N(S) = (n - |S|)!$  for any  $S \subseteq [n]$ .

Indeed,  $\pi \in X$  has all properties with indices from  $S$  if and only if all  $i \in S$  are fixed points of  $\pi$ . On the other elements, i.e. on  $[n] \setminus S$ ,  $\pi$  may be an arbitrary bijection, so there are  $(n - |S|)!$  choices for  $\pi$ .

## Derangements [2]

3. Using Theorem 14, the number of derangements is given by

$$\begin{aligned} |X \setminus (X_1 \cup \dots \cup X_n)| &\stackrel{\text{PIE}}{=} \sum_{S \subseteq [n]} (-1)^{|S|} |N(S)| \\ &= \sum_{S \subseteq [n]} (-1)^{|S|} (n - |S|)! \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i} (n - i)! \end{aligned}$$

In the last step, we used that  $(-1)^{|S|} (n - |S|)!$  only depends on the size of  $S$ , and there are  $\binom{n}{i}$  sets  $S \subseteq [n]$  of size  $i$ .

□

## Derangements [3]

**Hat-check problem:**  $n$  guests check their hats; the attendant loses the tickets.

As  $n \rightarrow \infty$ , the probability that no guest gets their own hat back converges to

$$\lim_{n \rightarrow \infty} \frac{|D_n|}{n!} = \sum_{i=0}^{\infty} \frac{(-1)^i}{i!} = \frac{1}{e} \approx 36.8\%$$

This probability barely changes for  $n \geq 5$ .

# Generating Functions

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*“A generating function is a clothesline on which we hang up a sequence of numbers for display.”*

*— Herbert Wilf, generatingfunctionology*

## Generating Functions

*A generating function is a device somewhat similar to a bag. Instead of carrying many little objects detachedly, which could be embarrassing, we put them all in a bag, and then we have only one object to carry, the bag.*

— George Pólya, Mathematics and Plausible Reasoning [1]

Encode a sequence  $(a_n)$  as coefficients of a formal power series  $G(x) = \sum_{n=0}^{\infty} a_n x^n$ .

Then manipulate  $G(x)$  *algebraically* (multiply, differentiate, compose) to derive counting identities and closed forms.

## Counting with Polynomials

**Problem.** Two fair dice. How many outcomes give sum  $n$ ?

Encode one die as a polynomial: the term  $x^k$  stands for “rolling  $k$  is possible.”

$$D(x) = x + x^2 + x^3 + x^4 + x^5 + x^6$$

Two independent dice  $\rightarrow$  multiply:

$$D(x)^2 = x^2 + 2x^3 + 3x^4 + 4x^5 + 5x^6 + \underline{6x^7} + 5x^8 + 4x^9 + 3x^{10} + 2x^{11} + x^{12}$$

The coefficient of  $x^n$  gives the count for sum  $n$ . For sum 7:  $[x^7]D(x)^2 = 6$ . ✓

One multiplication solves the problem for *every*  $n$  simultaneously.

This polynomial is a *generating function* — now let's understand what they are and why this works.

# Ordinary Generating Functions

**Definition 23:** An *ordinary generating function* (OGF) of a sequence  $a_n$  is a *power series*

$$G(a_n; x) = \sum_{n=0}^{\infty} a_n x^n$$

*Example:* The *sequence*  $a_n = (a_0, a_1, a_2, \dots)$  is *generated* by the OGF  $G(x) = a_0 + a_1x + a_2x^2 + \dots$

*Example:*  $G(x) = 3 + 8x^2 + x^3 + \frac{1}{7}x^5 + 100x^6 + \dots$  *generates* the sequence  $(3, 0, 8, 1, 0, \frac{1}{7}, 100, 0, \dots)$

*Example:* Consider a long division of 1 by  $(1 - x)$ , the result is an infinite power series

$$\frac{1}{1-x} = 1 + x^1 + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n$$

Note that all coefficients are 1. Thus, the generating function of  $(1, 1, 1, \dots)$  is  $G(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ .

## The Core Generating Function

Another proof that  $(1, 1, 1, \dots)$  is generated by  $G(x) = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x} = S$ :

$$S = 1 + x + x^2 + x^3 + \dots$$

$$x \cdot S = x + x^2 + x^3 + \dots$$

---

$$S - x \cdot S = 1$$

$$\text{Thus, } S = \frac{1}{1-x}.$$

The generating function  $G(x) = 1 + x + x^2 + \dots$  is also known as the *Maclaurin series* of  $\frac{1}{1-x}$ .

## More Examples of Generating Functions

| Formula          | Power series                                                     | Sequence              | Description          |
|------------------|------------------------------------------------------------------|-----------------------|----------------------|
| $\frac{1}{1-x}$  | $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + \dots$            | $(1, 1, 1, \dots)$    | constant 1           |
| $\frac{2}{1-x}$  | $\sum_{n=0}^{\infty} 2x^n = 2 + 2x + 2x^2 + 2x^3 + \dots$        | $(2, 2, 2, \dots)$    | constant 2           |
| $\frac{x}{1-x}$  | $\sum_{n=1}^{\infty} x^n = 0 + x + x^2 + x^3 + \dots$            | $(0, 1, 1, 1, \dots)$ | right shift          |
| $\frac{1}{1+x}$  | $\sum_{n=0}^{\infty} (-1)^n x^n = 0 + 1 - x + x^2 - x^3 + \dots$ | $(1, -1, 1, \dots)$   | sign-alternating 1's |
| $\frac{1}{1-3x}$ | $\sum_{n=0}^{\infty} 3^n x^n = 1 + 3x + 9x^2 + 27x^3 + \dots$    | $(1, 3, 9, \dots)$    | powers of 3          |

## More Examples of Generating Functions [2]

| Formula             | Power series                                                  | Sequence              | Description     |
|---------------------|---------------------------------------------------------------|-----------------------|-----------------|
| $\frac{1}{1-x^2}$   | $\sum_{n=0}^{\infty} x^{2n} = 1 + x^2 + x^4 + x^6 + \dots$    | $(1, 0, 1, 0, \dots)$ | regular gaps    |
| $\frac{1}{(1-x)^2}$ | $\sum_{n=0}^{\infty} (n+1)x^n = 1 + 2x + 3x^2 + 4x^3 + \dots$ | $(1, 2, 3, 4, \dots)$ | natural numbers |

$$\begin{aligned}
 \frac{1-x^{n+1}}{1-x} &= \frac{1}{1-x} - \frac{x^{n+1}}{1-x} = \\
 &\triangleq (1, 1, 1, \dots) - \underbrace{(0, 0, \dots, 0, 1, 1, \dots)}_{n+1 \text{ zeros}} = \\
 &= \underbrace{(1, 1, \dots, 1, 0, 0, \dots)}_{n+1 \text{ ones}} = \\
 &\triangleq 1 + x + x^2 + \dots + x^n
 \end{aligned}$$

## Exercises

*Example:* Find the GF for odd numbers:  $(1, 3, 5, 7, \dots)$ .

**Hint:** write  $A(x) = 1 + 3x + 5x^2 + 7x^3 + \dots$ , then compute  $x \cdot A(x)$  and subtract from  $A(x)$ .

What familiar series do you see in the result?

*Example:* Find the GF for  $(1, 3, 7, 15, 31, 63, \dots)$ , which satisfies  $a_n = 3a_{n-1} - 2a_{n-2}$  with  $a_0 = 1, a_1 = 3$ .

**Hint:** write  $A(x)$ ,  $-3xA(x)$ , and  $+2x^2A(x)$  stacked and add them column by column.

Most terms cancel because of the recurrence. What remains?

## Solutions

*Example: Odd numbers.* Let  $A(x) = 1 + 3x + 5x^2 + 7x^3 + \dots$

$$\begin{array}{r} A(x) = 1 + 3x + 5x^2 + 7x^3 + 9x^4 + \dots \\ x \cdot A(x) = \phantom{1} + x + 3x^2 + 5x^3 + 7x^4 + \dots \\ \hline A - x \cdot A = 1 + 2x + 2x^2 + 2x^3 + 2x^4 + \dots \end{array}$$

The result is  $1 + 2(x + x^2 + x^3 + \dots) = 1 + 2 \cdot \frac{x}{1-x}$ , so:

$$A(x) = \frac{1 + 2 \cdot \frac{x}{1-x}}{1-x} = \frac{1-x+2x}{(1-x)^2} = \frac{1+x}{(1-x)^2}$$

Check:  $\frac{1+x}{(1-x)^2} = (1, 1, 1, \dots) + (0, 2, 4, 6, 8, \dots) = (1, 3, 5, 7, 9, \dots)$  ✓

## Solutions [2]

*Example: Recurrence sequence.* Stack  $A$ ,  $-3x \cdot A$ ,  $+2x^2 \cdot A$  and add:

$$\begin{array}{rcl} A(x) & = & 1 + 3x + 7x^2 + 15x^3 + 31x^4 + \dots \\ -3x \cdot A(x) & = & -3x - 9x^2 - 21x^3 - 45x^4 - \dots \\ 2x^2 \cdot A(x) & = & 2x^2 + 6x^3 + 18x^4 + \dots \\ \hline (1 - 3x + 2x^2) \cdot A(x) & = & 1 \end{array}$$

Every column beyond  $x^0$  vanishes because  $a_n - 3a_{n-1} + 2a_{n-2} = 0$  by the recurrence.

$$\text{Thus, } A(x) = \frac{1}{1 - 3x + 2x^2}$$

# Solving Combinatorial Problems via Generating Functions

*Example:* Find the number of integer solutions to  $y_1 + y_2 + y_3 = 12$  with  $0 \leq y_i \leq 6$ .

- Possible values for  $y_1$  are  $0 \leq y_1 \leq 6$ .
  - ▶ There is a *single* way to select  $y_1 = 0$ . The same for other values among  $1, \dots, 6$ .
  - ▶ There are *no* ways to select any value of  $y_1$  higher than 6.
  - ▶ The *number of ways to select  $y_1$  to be equal to  $n$*  forms a sequence  $(1, 1, 1, 1, 1, 1, 0, \dots)$ .
  - ▶ Write this sequence as a polynomial  $x^0 + x^1 + \dots + x^6$ .
  - ▶ Do the same for  $y_2$  and  $y_3$  (*in isolation!*).
- Since all combinations of  $y_1, y_2$  and  $y_3$  are valid non-conflicting solutions, we can multiply those polynomials and obtain the *generating function*  $G(x) = (1 + x + x^2 + \dots + x^6)^3$ .
  - ▶ For each  $n$ , the coefficient of  $x^n$  in  $G(x)$  is the number of integer solutions to  $y_1 + y_2 + y_3 = n$ .
  - ▶ In particular, we are interested in the coefficient of  $x^{12}$  in  $G(x)$ , denoted  $[x^{12}]G(x)$ .
  - ▶ Use ~~pen and paper~~ Wolfram Alpha to expand  $G(x)$ :

$$G(x) = x^{18} + 3x^{17} + 6x^{16} + \dots + \underline{28x^{12}} + \dots + 6x^2 + 3x + 1$$

- The *answer* is  $[x^{12}]G(x) = 28$  solutions.

## Slightly More Complex Combinatorial Problem

*Example:* Suppose we have marbles of three different colors (●, ●, ●), and we want to *count* the number of ways to select 20 marbles, such that:

- There are an even number of ●:  $1 + x^2 + x^4 + \dots + x^{20}$ .
- There are at least 12 ●:  $x^{12} + x^{13} + \dots + x^{20}$ .
- There are at most 5 ●:  $1 + x + x^2 + x^3 + x^4 + x^5$ .

Multiply polynomials and find  $[x^{20}]G(x)$ :

$$\begin{aligned} & [x^{20}](1 + x^2 + x^4 + \dots + x^{20})(x^{12} + x^{13} + \dots + x^{20})(1 + x + x^2 + x^3 + x^4 + x^5) = \\ & = [x^{20}](x^{45} + 2x^{44} + \dots + \underline{21x^{20}} + \dots + 2x^{13} + x^{12}) \\ & = 21 \end{aligned}$$

## Using Power Series in Combinatorial Problems

*Example:* Find the number of integer solutions to  $a_1 + a_2 + a_3 = 12$  with  $a_1 \geq 2, 3 \leq a_2 \leq 6, 0 \leq a_3 \leq 9$ .

- Compose the generating function:

$$G(x) = (x^2 + x^3 + \dots) \cdot (x^3 + x^4 + x^5 + x^6) \cdot (1 + x + x^2 + \dots + x^9)$$

- Substitute the power series with the corresponding simple forms:

$$G(x) = \left(x^2 \cdot \frac{1}{1-x}\right) \cdot \left(x^3 \cdot \frac{1-x^4}{1-x}\right) \cdot \left(\frac{1-x^{10}}{1-x}\right)$$

- Expand the series:

$$G(x) = x^5 + 3x^6 + 6x^7 + 10x^8 + 14x^9 + 18x^{10} + 22x^{11} + \underline{26x^{12}} + 30x^{13} + \\ 34x^{14} + 37x^{15} + 39x^{16} + 40x^{17} + \dots + 40x^n + \dots$$

- Sequence:  $(g_n) = (0, 0, 0, 0, 0, 1, 3, 6, 10, 14, 18, 22, 26, 30, 34, 37, 39, \overline{40}, \dots)$
- Answer for  $n = 12$  is  $[x^{12}]G(x) = 26$ .

## Operations on Generating Functions

Let  $F(x) = \sum_{n=0}^{\infty} a_n x^n$  and  $G(x) = \sum_{n=0}^{\infty} b_n x^n$  be ordinary generating functions.

All operations are performed *term-wise* on the power series.

| Operation                                                                                 | Result                                                                              |
|-------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Differentiate $F(x)$ term-wise                                                            | $F'(x) = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n$                                     |
| Multiply $F(x)$ by a scalar $\lambda \in \mathbb{R}$ term-wise                            | $\lambda F(x) = \sum_{n=0}^{\infty} \lambda a_n x^n$                                |
| Add $F(x)$ and $G(x)$ term-wise                                                           | $F(x) + G(x) = \sum_{n=0}^{\infty} (a_n + b_n) x^n$                                 |
| Multiply $F(x)$ and $G(x)$ term-wise<br>( <i>Cauchy product</i> , or <i>convolution</i> ) | $F(x) \cdot G(x) = \sum_{n=0}^{\infty} \left( \sum_{k=0}^n a_k b_{n-k} \right) x^n$ |

## Catalan Numbers via Generating Functions

So far, we *composed* generating functions from known polynomials and read off coefficients.

Now we turn the idea around:

1. Start from a *recursive counting argument*.
2. Derive a *functional equation* for  $F(x)$ .
3. Solve it.

This requires a heavier tool — *Newton's binomial theorem* — but the payoff is one of the most famous sequences in combinatorics.

**Historical note:** Eugène Catalan (1814–1894) studied these numbers in the context of polygon triangulation. They appear in dozens of seemingly unrelated counting problems — a phenomenon that generating functions help explain.

## Well-Formed Parenthesis Expressions

*Example:* Find the number of *well-formed parenthesis expressions* with  $n$  pairs of parenthesis.

For example, “( ( ) ( ) )” is a well-formed parenthesis expression with 4 pairs of parenthesis.

Formally, a permutation of the multiset  $\{n \cdot "(", n \cdot ")"\}$  is *well-formed* if reading it from left to right and counting “+1” for every opening parenthesis “(“ and “−1” for every closing parenthesis “)” never yields a negative number at any time.

Every well-formed expression with  $n \geq 1$  pairs of parenthesis starts with “(“ and there is a unique matching “)” such that the sequence in between and the sequence after are well-formed. For example:

$$()()() \quad (())() \quad ()(())()$$

In other words, a well-formed expression with  $n$  pairs of parenthesis is obtained by putting a well-formed expression with  $k$  pairs in between “(“ and “)” and then appending a well-formed expression with  $n - k - 1$  pairs of parenthesis. This gives the equation:

$$a_n = \sum_{k=0}^{n-1} a_k a_{n-k-1}$$

## Well-Formed Parenthesis Expressions [2]

Let  $F(x)$  be a generating function for  $a_n$ , then we know:

$$\begin{aligned} F(x) &= \sum_{n=0}^{\infty} a_n x^n = 1 + \sum_{n=1}^{\infty} \left( \sum_{k=0}^{n-1} a_k a_{n-k-1} \right) x^n = 1 + \sum_{n=0}^{\infty} \left( \sum_{k=0}^n a_k a_{n-k} \right) x^{n+1} \\ &= 1 + x \cdot \sum_{n=0}^{\infty} \left( \sum_{k=0}^n a_k a_{n-k} \right) x^n = 1 + x \cdot F(x)^2 \end{aligned}$$

Solving the equation  $xF^2 - F + 1 = 0$  for  $F$  gives:

$$F(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$$

We are only interested in the root with the *minus* sign, since  $\lim_{x \rightarrow 0} F(x) = a_0 = 1$ :

$$\lim_{x \rightarrow 0} F(x) = \lim_{x \rightarrow 0} \frac{1 - \sqrt{1-4x}}{2x} = \lim_{x \rightarrow 0} \frac{\frac{2}{\sqrt{1-4x}}}{2} = 1$$

## Newton's Binomial Theorem

To extract the coefficients of  $F(x) = \frac{1-\sqrt{1-4x}}{2x}$ , we need to expand  $\sqrt{1-4x}$  as a power series.

This requires extending the binomial theorem to *real* exponents.

Let's revisit the binomial theorem:

$$(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k = \sum_{k=0}^{\infty} \binom{n}{k} x^k \quad \forall n \in \mathbb{N}$$

where  $\binom{n}{k} = 0$  for  $k > n$ .

**Note:** This shows that  $(1+x)^n$  is the generating function for the series  $(a_k)_{k \in \mathbb{N}}$  with  $a_k = \binom{n}{k}$ .

We can extend this result from natural numbers  $n \in \mathbb{N}$  to any *real* number  $n \in \mathbb{R}$ .

## Binomial Coefficients for Real Numbers

**Definition 24:** Let  $p(n, k) = n \cdot (n - 1) \cdot \dots \cdot (n - k + 1)$ , also called the *falling factorial*  $n^{\underline{k}}$ .

Extend the definition of binomial coefficients for real numbers  $n, k \in \mathbb{R}$ :

$$\binom{n}{k} = \frac{p(n, k)}{k!}$$

**Note:** This definition aligns with the definition of binomial coefficients for natural numbers:

$$\binom{n}{k} = \frac{n!}{k! \cdot (n - k)!} = \frac{n \cdot (n - 1) \cdot \dots \cdot (n - k + 1)}{k!}$$

*Example:* Consider the number “ $-7/2$  choose 5”:

$$\binom{-7/2}{5} = \frac{-7/2 \cdot -9/2 \cdot -11/2 \cdot -13/2 \cdot -15/2}{5!} = -\frac{9009}{256} \approx -35$$

**Note:**  $p(n, 0) = 1$  and for  $k \geq 1$ , we have  $p(n, k) = (n - k + 1) \cdot p(n, k - 1) = n \cdot p(n - 1, k - 1)$ . (★)

## Extended Newton's Binomial Theorem

**Theorem 17:** For all non-zero  $n \in \mathbb{R}$ , we have:

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k$$

*Example:* Let  $n = 1/2$ , then we have an identity for  $\sqrt{1+x}$ :

$$\sqrt{1+x} = \sum_{k=0}^{\infty} \binom{1/2}{k} x^k$$

To actually *use* this fact, we need some *lemma*...

**Lemma 18:** For any integer  $n \geq 1$ , we have:

$$\binom{1/2}{n} = (-1)^{n+1} \cdot \binom{2n-2}{n-1} \cdot \frac{1}{2^{2n-1}} \cdot \frac{1}{n}$$

## Extended Newton's Binomial Theorem [2]

**Proof:** By induction on  $n$ .

**Base:**  $n = 1$ .

$$\binom{1/2}{1} = \frac{1/2}{1!} = \frac{1}{2} = 1 \cdot 1 \cdot \frac{1}{2} \cdot 1 = \underbrace{(-1)^2}_1 \cdot \underbrace{\binom{2-2}{1-1}}_1 \cdot \underbrace{\frac{1}{2^{2-1}}}_{\frac{1}{2}} \cdot \underbrace{\frac{1}{1}}_1$$

**Induction step:**  $n$  to  $n + 1$  for  $n > 1$ . We use the recursion  $(\star) p(n, k) = n \cdot p(n - 1, k - 1)$ :

$$\begin{aligned} \binom{1/2}{n+1} &= \frac{p(1/2, n+1)}{(n+1)!} = \frac{(1/2 - (n+1) + 1) \cdot p(1/2, n)}{(n+1) \cdot n!} = -\frac{n-1/2}{n+1} \binom{1/2}{n} \\ &\stackrel{\text{IH}}{=} -\frac{n-1/2}{n+1} (-1)^{n+1} \cdot \binom{2n-2}{n-1} \cdot \frac{1}{2^{2n-1}} \cdot \frac{1}{n} \\ &= \frac{2n}{2n} \cdot \frac{2n-1}{2n} \cdot (-1)^{n+2} \cdot \binom{2n-2}{n-1} \cdot \frac{1}{2^{2n-1}} \cdot \frac{1}{n+1} \\ &= (-1)^{n+2} \cdot \underbrace{\frac{(2n-2)! \cdot (2n-1) \cdot (2n)}{(n-1)! \cdot (n-1) \cdot n \cdot n}}_{\binom{2n}{n}} \cdot \frac{1}{2^{2n+1}} \cdot \frac{1}{n+1} \end{aligned} \quad \square$$

## Catalan Numbers

**Proposition 19:** Now we can expand  $\sqrt{1+n}$  into the following series:

$$\sqrt{1+n} = \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \sum_{n=1}^{\infty} -2 \cdot \binom{2n-2}{n-1} \cdot (-1)^n \cdot \frac{1}{2^{2n}} \cdot \frac{1}{n} \cdot x^n$$

*Example:* Going back to the example with the number of well-formed paranthesis expressions, we get:

$$\begin{aligned} F(x) &= \frac{1 - \sqrt{1-4x}}{2x} = \frac{1}{2x} \sum_{n=1}^{\infty} 2 \cdot \binom{2n-2}{n-1} \cdot (-1)^n \cdot \frac{1}{2^{2n}} \cdot \frac{1}{n} \cdot (-4x)^n \\ &= \frac{1}{x} \sum_{n=1}^{\infty} \binom{2n-2}{n-1} \frac{1}{n} x^n = \sum_{n=0}^{\infty} \binom{2n}{n} \frac{1}{n+1} x^n \end{aligned}$$

The numbers  $C_n := \binom{2n}{n} \frac{1}{n+1}$  are called *Catalan numbers*.

## Catalan Numbers [2]

**Catalan numbers**  $C_n = \frac{1}{n+1} \binom{2n}{n}$  count many combinatorial structures:

- well-formed parenthesis strings of length  $2n$ ,
- full binary trees with  $n + 1$  leaves,
- triangulations of a convex  $(n + 2)$ -gon,
- monotone lattice paths from  $(0, 0)$  to  $(n, n)$  not crossing the diagonal.

$$C_0 = 1, \quad C_1 = 1, \quad C_2 = 2, \quad C_3 = 5, \quad C_4 = 14, \quad C_5 = 42, \quad \dots$$

# Recurrence Relations

---

*“A certain man placed one pair of rabbits in a place enclosed on all sides by a wall. How many pairs of rabbits can be produced from that pair in one year?”*

*— Leonardo Fibonacci, Liber Abaci (1202)*

## Recurrence Relations

A *recurrence relation* defines a sequence by expressing each term via its predecessors.

Solving one means finding a *closed-form* formula that avoids recursion.

*Example:* *Arithmetic progression* as a recurrence:

$$a_n = \begin{cases} a_0 = \text{const} & \text{if } n = 0 \\ a_{n-1} + d & \text{if } n > 0 \end{cases}$$

Solution by telescoping:  $a_n = a_0 + n \cdot d$ .

## Recursive Structure

Many combinatorial objects have *recursive structure*: a large instance decomposes into smaller instances of the same type.

A *recurrence* is a formula that captures this decomposition.

*Example*: How many binary strings of length  $n$  contain *no two consecutive* 1s? Let  $B(n)$  denote the count.

Look at the last bit of a valid string of length  $n$ :

- **Ends in 0**: the first  $n - 1$  bits can be any valid string, that gives  $B(n - 1)$  options.
- **Ends in 1**: then the bit before it *must* be 0, so the first  $n - 2$  bits can be any valid string, giving  $B(n - 2)$  options.

Therefore  $B(n) = B(n - 1) + B(n - 2)$ , with  $B(1) = 2$  and  $B(2) = 3$ .

This is the *Fibonacci recurrence* with shifted initial conditions.

## Telescoping

Before reaching for characteristic equations, try the simplest approach: *unwind the recurrence*.

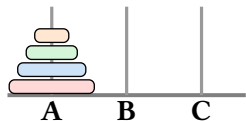
*Example:* Solve  $T(n) = T(n - 1) + n$  with  $T(0) = 0$ .

Unwind by substitution:

$$\begin{aligned}T(n) &= T(n - 1) + n \\&= T(n - 2) + (n - 1) + n \\&= \vdots \\&= T(0) + 1 + 2 + \dots + n \\&= \frac{n(n + 1)}{2}\end{aligned}$$

**Theorem 20:** If  $a_n = a_{n-1} + f(n)$ , then  $a_n = a_0 + \sum_{i=1}^n f(i)$  (telescoping sum).

## Tower of Hanoi



Move  $n$  disks from peg A to peg C, one at a time, never placing a larger disk on a smaller one.

**Strategy:** move the top  $n - 1$  disks to peg B, move the largest disk to peg C, then move the  $n - 1$  disks from B to C.

*Example:* Let  $T(n)$  be the minimum number of moves. Recurrence:  $T(n) = 2T(n - 1) + 1$ , with  $T(1) = 1$ .

Unwind:

$$\begin{aligned}T(n) &= 2T(n - 1) + 1 \\&= 2(2T(n - 2) + 1) + 1 = 4T(n - 2) + 3 \\&= 8T(n - 3) + 7 \\&= 2^k T(n - k) + (2^k - 1)\end{aligned}$$

Setting  $k = n - 1$ :  $T(n) = 2^{n-1}T(1) + 2^{n-1} - 1 = 2^n - 1$ .

# Linear Homogeneous Recurrence Relations

**Definition 25:** A *linear homogeneous* recurrence relation *of degree  $k$*  with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k},$$

where  $c_1, c_2, \dots, c_k$  are constants (real or complex numbers), and  $c_k \neq 0$ .

*Examples:*

- $b_n = 2.71b_{n-1}$  is a linear homogeneous recurrence relation of degree 1.
- $F_n = F_{n-1} + F_{n-2}$  is a linear homogeneous recurrence relation of degree 2.
- The recurrence relation  $a_n = a_{n-1} + a_{n-2}^2$  is *not linear*.
- The recurrence relation  $H_n = 2H_{n-1} + 1$  is *not homogeneous*.

## Characteristic Equations

Fix a linear homogeneous recurrence of degree  $k$ :

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} \quad (\star)$$

If we try a solution of the form  $a_n = r^n$  (with  $r \neq 0$ ), substituting into  $(\star)$  and dividing by  $r^{n-k}$  yields:

**Definition 26:** The *characteristic equation* of  $(\star)$  is:

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} - \dots - c_k = 0$$

Its solutions  $r_1, r_2, \dots$  are the *characteristic roots* of the recurrence. Each root yields a solution  $a_n = r^n$ .

**Why guess  $r^n$ ?** Plug  $a_n = r^n$  into the recurrence:

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$$

Divide by  $r^{n-k}$  (assuming  $r \neq 0$ ) and you get exactly the characteristic equation. So  $r^n$  is a solution precisely when  $r$  is a root, and by linearity any linear combination of such solutions works too.

## Distinct Roots Case

**Theorem 21:** Let  $c_1$  and  $c_2$  be real numbers. Suppose that  $r^2 - c_1r - c_2 = 0$  has two *distinct* roots  $r_1$  and  $r_2$ . Then the sequence  $(a_n)$  is a solution of the recurrence relation  $a_n = c_1a_{n-1} + c_2a_{n-2}$  if and only if  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$  for  $n = 0, 1, 2, \dots$ , where  $\alpha_1$  and  $\alpha_2$  are constants.

**Proof:** We prove both directions.

( $\Rightarrow$ ) **Suppose**  $a_n = \alpha_1r_1^n + \alpha_2r_2^n$ . We verify it satisfies  $a_n = c_1a_{n-1} + c_2a_{n-2}$ . Since  $r_i$  is a root of  $r^2 - c_1r - c_2 = 0$ , we have  $r_i^2 = c_1r_i + c_2$ . Multiply both sides by  $r_i^{n-2}$  to get  $r_i^n = c_1r_i^{n-1} + c_2r_i^{n-2}$ . This holds for  $i = 1$  and  $i = 2$  separately, so taking a linear combination preserves equality:

$$\begin{aligned}c_1a_{n-1} + c_2a_{n-2} &= c_1(\alpha_1r_1^{n-1} + \alpha_2r_2^{n-1}) + c_2(\alpha_1r_1^{n-2} + \alpha_2r_2^{n-2}) \\&= \alpha_1(c_1r_1^{n-1} + c_2r_1^{n-2}) + \alpha_2(c_1r_2^{n-1} + c_2r_2^{n-2}) \\&= \alpha_1r_1^n + \alpha_2r_2^n = a_n.\end{aligned}$$

## Distinct Roots Case [2]

( $\Leftarrow$ ) **Suppose  $a_n$  satisfies the recurrence.** We show  $a_n$  *must* be of the form  $\alpha_1 r_1^n + \alpha_2 r_2^n$ .

The recurrence and initial conditions  $a_0, a_1$  determine the entire sequence: each  $a_n$  is computed from  $a_{n-1}$  and  $a_{n-2}$ . So it suffices to find  $\alpha_1, \alpha_2$  such that  $\alpha_1 r_1^n + \alpha_2 r_2^n$  matches the two initial values:

$$\begin{cases} \alpha_1 + \alpha_2 = a_0 \\ \alpha_1 r_1 + \alpha_2 r_2 = a_1 \end{cases}$$

This is a  $2 \times 2$  linear system with determinant  $r_2 - r_1$ . Since  $r_1 \neq r_2$ , the determinant is nonzero, so a unique solution  $(\alpha_1, \alpha_2)$  exists. The sequence  $\alpha_1 r_1^n + \alpha_2 r_2^n$  matches the initial conditions and satisfies the recurrence (by the first direction), so it *equals*  $a_n$  by uniqueness.  $\square$

## Worked Example

*Example:* Solve  $a_n = a_{n-1} + 2a_{n-2}$  with  $a_0 = 2$  and  $a_1 = 7$ .

- The *characteristic* equation is  $r^2 - r - 2 = 0$ .
- It has two *distinct* roots  $r_1 = 2$  and  $r_2 = -1$ .
- The sequence  $(a_n)$  is a solution iff  $a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$  for  $n = 0, 1, 2, \dots$  and some constants  $\alpha_1$  and  $\alpha_2$ .

$$\begin{cases} a_0 = 2 = \alpha_1 + \alpha_2 \\ a_1 = 7 = \alpha_1 \cdot 2 + \alpha_2 \cdot (-1) \end{cases}$$

- Solving these two equations gives  $\alpha_1 = 3$  and  $\alpha_2 = -1$ .
- Hence, the *solution* to the recurrence equation with given initial conditions is the sequence  $(a_n)$  with

$$a_n = 3 \cdot 2^n - (-1)^n$$

## Fibonacci Numbers

*Example:* The recurrence  $F_n = F_{n-1} + F_{n-2}$  has characteristic equation  $r^2 - r - 1 = 0$  with roots:

$$\varphi = \frac{1 + \sqrt{5}}{2} \approx 1.618, \quad \psi = \frac{1 - \sqrt{5}}{2} \approx -0.618.$$

The general solution is  $F_n = \alpha_1 \varphi^n + \alpha_2 \psi^n$  for constants  $\alpha_1, \alpha_2$ .

Using initial conditions  $F_0 = 0$  and  $F_1 = 1$ :

$$\begin{cases} \alpha_1 + \alpha_2 = 0 \\ \alpha_1 \varphi + \alpha_2 \psi = 1 \end{cases}$$

## Binet's Formula

Solving for  $\alpha_1, \alpha_2$  yields  $\alpha_1 = 1/\sqrt{5}$  and  $\alpha_2 = -1/\sqrt{5}$ , giving *Binet's formula*:

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}$$

Since  $|\psi| \approx 0.618 < 1$ , the term  $\psi^n \rightarrow 0$ , so  $F_n$  is the nearest integer to  $\varphi^n/\sqrt{5}$ .

In particular,  $F_{n+1}/F_n \rightarrow \varphi$  as  $n \rightarrow \infty$ .

| $n$           | 1     | 2     | 3     | 4     | 5     | 6     | 7     | 8     |
|---------------|-------|-------|-------|-------|-------|-------|-------|-------|
| $F_n$         | 1     | 1     | 2     | 3     | 5     | 8     | 13    | 21    |
| $F_{n+1}/F_n$ | 1.000 | 2.000 | 1.500 | 1.667 | 1.600 | 1.625 | 1.615 | 1.619 |

## Eigenvalues and the Companion Matrix

**Definition 27:** Let  $A$  be an  $m \times m$  matrix. A scalar  $\lambda$  is an *eigenvalue* of  $A$  if there exists a nonzero vector  $\vec{v}$  (called an *eigenvector*) such that

$$A\vec{v} = \lambda\vec{v}$$

The eigenvalues are the roots of the *characteristic polynomial*  $p(\lambda) = \det(A - \lambda I)$ .

Why does this matter for recurrences? Define the *state vector*  $\vec{v}_n = (a_n, a_{n-1}, \dots, a_{n-k+1})^T$ . Then the recurrence  $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$  is equivalent to  $\vec{v}_n = C\vec{v}_{n-1}$ , where  $C$  is the *companion matrix*:

$$C = \begin{pmatrix} c_1 & c_2 & c_3 & \dots & c_k \\ 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}$$

## Eigenvalues and the Companion Matrix [2]

Iterating gives  $\vec{v}_n = C^n \vec{v}_0$ , so computing  $a_n$  reduces to *matrix exponentiation*. The characteristic polynomial of  $C$  is exactly  $p(\lambda) = \lambda^k - c_1 \lambda^{k-1} - \dots - c_k$ , i.e. the same characteristic equation we solve for the recurrence.

For  $k = 2$ :  $C = \begin{pmatrix} c_1 & c_2 \\ 1 & 0 \end{pmatrix}$ , and  $\det(C - \lambda I) = \lambda^2 - c_1 \lambda - c_2$ .

*Example:* Fibonacci:  $C = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , eigenvalues  $\varphi, \psi$ . Diagonalizing  $C = PDP^{-1}$  yields  $C^n = PD^nP^{-1}$ , recovering Binet's formula. Matrix exponentiation computes  $F_n$  in  $O(\log n)$  multiplications (binary exponentiation), vs.  $O(n)$  for naive iteration.

## Single (Repeated) Root Case

**Theorem 22:** Let  $c_1$  and  $c_2$  be real numbers. Suppose that  $r^2 - c_1r - c_2 = 0$  has a *single (repeated)* root  $r_0$ . A sequence  $(a_n)$  is a solution of the recurrence relation  $a_n = c_1a_{n-1} + c_2a_{n-2}$  if and only if  $a_n = (\alpha_1 + \alpha_2n)r_0^n$  for  $n = 0, 1, 2, \dots$ , where  $\alpha_1$  and  $\alpha_2$  are constants.

**Why  $nr_0^n$ ?** When the root is repeated,  $r_0^n$  alone only provides one linearly independent solution. The second solution  $nr_0^n$  arises by taking the *derivative* of  $r^n$  with respect to  $r$  and then evaluating at  $r = r_0$ :

$$\frac{d}{dr}(r^n) = nr^{n-1} \implies \frac{d}{dr}(r^n) \cdot r = nr^n$$

This is analogous to the resonance term  $te^{\lambda t}$  in differential equations.

## Single (Repeated) Root Case [2]

*Example:* Solve  $a_n = 6a_{n-1} - 9a_{n-2}$  with  $a_0 = 1$  and  $a_1 = 6$ .

The characteristic equation is  $r^2 - 6r + 9 = 0$  with a single (repeated) root  $r_0 = 3$ .

Hence, the solution is of the form  $a_n = (\alpha_1 + \alpha_2 n) 3^n$ .

$$\begin{cases} a_0 = 1 = \alpha_1 \\ a_1 = 6 = (\alpha_1 + \alpha_2) \cdot 3 \end{cases} \implies \begin{cases} \alpha_1 = 1 \\ \alpha_2 = 1 \end{cases}$$

Thus, the *solution* is  $a_n = (1 + n) 3^n$ .

## General Case

**Theorem 23:** Let  $c_1, \dots, c_k$  be real numbers. Suppose that the recurrence  $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$  has  $t$  distinct characteristic roots  $r_1, \dots, r_t$  with multiplicities  $m_1, \dots, m_t$ , respectively, so that  $m_i \geq 1$  for  $i = 1, \dots, t$  and  $m_1 + \dots + m_t = k$ .

Then the general solution of the recurrence relation is given by

$$a_n = \sum_{i=1}^t \left( r_i^n \cdot \sum_{j=0}^{m_i-1} \alpha_{i,j} n^j \right)$$

where  $\alpha_{i,j}$  are constants for  $1 \leq i \leq t$  and  $0 \leq j \leq m_i - 1$ .

*Example:* Find generic solution for  $a_n = 7a_{n-1} - 16a_{n-2} + 12a_{n-3}$ .

The characteristic equation  $r^3 - 7r^2 + 16r - 12 = 0$  has  $t = 2$  distinct roots:  $r_1 = 2$  repeated  $m_1 = 2$  times, and  $r_2 = 3$  with multiplicity  $m_2 = 1$ . Hence, the solution is of the form

$$a_n = (\alpha_{1,0} + \alpha_{1,1}n) 2^n + \alpha_{2,0} 3^n$$

## Complex Roots

When the characteristic equation has complex roots, they come in *conjugate pairs*  $\rho e^{\pm i\theta}$  (for recurrences with real coefficients).

**Definition 28:** For conjugate roots  $\rho e^{\pm i\theta}$ , the real general solution is:

$$a_n = \rho^n (\alpha \cos(n\theta) + \beta \sin(n\theta))$$

This follows from Euler's formula:  $\rho^n e^{in\theta} = \rho^n (\cos(n\theta) + i \sin(n\theta))$ .

Complex roots produce *oscillating exponential* behavior: magnitude  $\rho^n$  grows (or decays), while  $\cos(n\theta)$  and  $\sin(n\theta)$  produce periodic oscillation with period  $2\pi/\theta$ .

## Complex Roots Example

*Example:* Solve  $a_n = 2a_{n-1} - 2a_{n-2}$  with  $a_0 = 1$  and  $a_1 = 1$ .

Characteristic equation  $r^2 - 2r + 2 = 0$ , roots  $r = 1 \pm i$ .

Polar form:  $1 + i = \sqrt{2}e^{i\pi/4}$ , so  $\rho = \sqrt{2}$  and  $\theta = \pi/4$ .

General solution:  $a_n = (\sqrt{2})^n (\alpha \cos(n\pi/4) + \beta \sin(n\pi/4))$ .

Initial conditions:  $a_0 = 1 \rightarrow \alpha = 1$  and  $a_1 = 1 \rightarrow \beta = 0$ .

Therefore  $a_n = (\sqrt{2})^n \cos(n\pi/4)$ .

## Root Types Summary

For  $a_n = c_1 a_{n-1} + c_2 a_{n-2}$ , the solution method is always the same four steps. Only the *form* depends on the roots.

| Step         | Distinct Real                     | Repeated                        | Complex                                             |
|--------------|-----------------------------------|---------------------------------|-----------------------------------------------------|
| 1. Char. eq. | $r^2 - c_1 r - c_2 = 0$           | $r^2 - c_1 r - c_2 = 0$         | $r^2 - c_1 r - c_2 = 0$                             |
| 2. Roots     | $r_1 \neq r_2$                    | $r_0$ (multiplicity 2)          | $\rho e^{\pm i\theta}$                              |
| 3. Form      | $\alpha_1 r_1^n + \alpha_2 r_2^n$ | $(\alpha_1 + \alpha_2 n) r_0^n$ | $\rho^n (\alpha \cos n\theta + \beta \sin n\theta)$ |
| 4. Fit ICs   | 2 equations                       | 2 equations                     | 2 equations                                         |

# Recurrences and Differential Equations

Linear recurrences are *discrete analogues* of linear differential equations. The entire theory mirrors the continuous case.

| Aspect          | Differential Equation            | Recurrence                           |
|-----------------|----------------------------------|--------------------------------------|
| 1st order       | $y' = cy \rightarrow y_0 e^{ct}$ | $a_n = ca_{n-1} \rightarrow a_0 c^n$ |
| 2nd order       | $y'' + py' + qy = 0$             | $a_n + pa_{n-1} + qa_{n-2} = 0$      |
| Char. eq.       | $\lambda^2 + p\lambda + q = 0$   | $r^2 + pr + q = 0$                   |
| Double root     | $(c_1 + c_2 t)e^{\lambda t}$     | $(\alpha_1 + \alpha_2 n)r_0^n$       |
| Particular sol. | Undetermined coefficients        | Same method                          |
| Resonance       | $te^{\lambda t}$ factors         | $nr_0^n$ factors                     |

# Solving Linear Recurrences

| Step            | Homogeneous                           | Non-homogeneous               |
|-----------------|---------------------------------------|-------------------------------|
| 1. Classify     | No $F(n)$ term?                       | Has $F(n)$ term?              |
| 2. Char. eq.    | $r^k - c_1 r^{k-1} - \dots - c_k = 0$ | Same (ignore $F(n)$ )         |
| 3. Roots        | Distinct / repeated / complex         | Same                          |
| 4. Homog. sol.  | $a_n^{(h)}$ from root type            | Same                          |
| 5. Partic. sol. | —                                     | Guess form matching $F(n)$    |
| 6. General      | $a_n = a_n^{(h)}$                     | $a_n = a_n^{(h)} + a_n^{(p)}$ |
| 7. Fit ICs      | Solve linear system                   | Solve linear system           |

## Particular solution cheat sheet:

- $F(n) = \text{polynomial of degree } d \implies \text{try } c_0 + c_1 n + \dots + c_d n^d$
- $F(n) = c \cdot r^n, r \text{ not a root} \implies \text{try } \beta r^n$

## Solving Linear Recurrences [2]

- $F(n) = c \cdot r^n$ ,  $r$  is a root (mult.  $m$ )  $\implies$  try  $\beta n^m r^n$  (resonance)

# Linear Non-Homogeneous Recurrence Relations

**Definition 29:** A *linear non-homogeneous* recurrence relation *of degree  $k$*  with constant coefficients is a recurrence relation of the form

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k} + F(n),$$

where  $c_1, c_2, \dots, c_k$  are constants and  $F(n) \neq 0$  is a function of  $n$  (the *forcing term*).

The *associated homogeneous recurrence* is the relation with  $F(n)$  removed:

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

*Example:*

- $a_n = 3a_{n-1} + 2n$  – non-homogeneous,  $F(n) = 2n$  (polynomial).
- $a_n = 2a_{n-1} + 3^n$  – non-homogeneous,  $F(n) = 3^n$  (exponential).
- $H_n = 2H_{n-1} + 1$  – non-homogeneous,  $F(n) = 1$  (constant).

## Solving Non-Homogeneous Recurrence Relations

**Theorem 24:** If  $(a_n^{(p)})$  is a *particular* solution of the non-homogeneous recurrence  $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k} + F(n)$ , then *every solution* is of the form  $a_n = a_n^{(p)} + a_n^{(h)}$ , where  $a_n^{(h)}$  is a solution of the associated homogeneous recurrence.

The recipe:

1. Solve the homogeneous part.
2. Find a particular solution by guessing a form matching  $F(n)$ .
3. Add them.
4. Fit initial conditions.

*Example:* Find all solutions of  $a_n = 3a_{n-1} + 2n$  with  $a_1 = 3$ .

*Homogeneous part:*  $a_n^{(h)} = \alpha 3^n$ .

*Particular solution:*  $F(n) = 2n$  is degree-1 polynomial, so try  $p_n = cn + d$ .

Substituting:  $cn + d = 3(c(n-1) + d) + 2n$ .

Equating coefficients:  $c = -1$ ,  $d = -\frac{3}{2}$ . So  $a_n^{(p)} = -n - \frac{3}{2}$ .

## Solving Non-Homogeneous Recurrence Relations [2]

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*General solution:*  $a_n = a_n^{(p)} + a_n^{(h)} = -n - \frac{3}{2} + \alpha 3^n$ .

With  $a_1 = 3$ :  $\alpha = \frac{11}{6}$ , giving  $a_n = -n - \frac{3}{2} + \left(\frac{11}{6}\right)3^n$ .

## Exponential Forcing Term

When  $F(n) = c \cdot r^n$ , try  $a_n^{(p)} = \beta r^n$ .

**Warning: resonance.** If  $r$  is already a characteristic root, the trial  $\beta r^n$  fails. Instead, multiply by  $n$ : try  $a_n^{(p)} = \beta n r^n$  (or  $n^m$  if  $r$  has multiplicity  $m$ ).

*Example:* Solve  $a_n = 2a_{n-1} + 3^n$  with  $a_0 = 1$ .

- *Homogeneous solution:*  $a_n^{(h)} = \alpha 2^n$ .
- *Particular solution.* Since 3 is *not* a characteristic root, try  $a_n^{(p)} = \beta \cdot 3^n$ :

$$\beta 3^n = 2\beta 3^{n-1} + 3^n \implies \beta = 2\frac{\beta}{3} + 1 \implies \beta = 3$$

So  $a_n^{(p)} = 3^{n+1}$ .

- *General solution:*  $a_n = \alpha 2^n + 3^{n+1}$ .
- Using  $a_0 = 1$ :  $1 = \alpha + 3$ , so  $\alpha = -2$ .
- The *solution* is  $a_n = 3^{n+1} - 2^{n+1}$ .

## Recurrences and Generating Functions

Both characteristic equations and generating functions solve the same recurrences, but differently.

**Key idea.** A generating function encodes the *entire sequence*  $(a_n)$  as coefficients of a power series  $A(x) = \sum a_n x^n$ . The recurrence turns into an *algebraic equation* for  $A(x)$ .

*Example:* *Derive* the generating function for Fibonacci:  $F_n = F_{n-1} + F_{n-2}$ ,  $F_0 = 0$ ,  $F_1 = 1$ .

Write  $G(x) = F_0 + F_1 x + F_2 x^2 + F_3 x^3 + \dots = x + x^2 + 2x^3 + 3x^4 + \dots$

Multiply by  $x$  and  $x^2$  to shift indices:

$$\begin{aligned} xG(x) &= F_0 x + F_1 x^2 + F_2 x^3 + \dots \\ x^2 G(x) &= F_0 x^2 + F_1 x^3 + \dots \end{aligned}$$

Since  $F_n = F_{n-1} + F_{n-2}$  for  $n \geq 2$ , each coefficient of  $G - xG - x^2 G$  vanishes except  $F_1 x = x$ :

$$G(x) - xG(x) - x^2 G(x) = x \implies G(x) = \frac{x}{1 - x - x^2}$$

## Recurrences and Generating Functions [2]

Now factor the denominator. The roots of  $1 - x - x^2 = 0$  are:

$$x = \frac{1 \pm \sqrt{5}}{-2} \implies x_1 = -\frac{1}{\varphi}, \quad x_2 = -\frac{1}{\psi}$$

where  $\varphi = \frac{1+\sqrt{5}}{2}$ ,  $\psi = \frac{1-\sqrt{5}}{2}$ .

The denominator  $1 - x - x^2$  and the characteristic polynomial  $r^2 - r - 1$  differ only by the substitution  $x \rightarrow \frac{1}{r}$ . Setting the denominator to zero gives values of  $x$  where the series *diverges* — these are the *poles* of  $G(x)$ . The characteristic roots  $r_1, r_2$  are the *reciprocals* of these poles.

## Recurrences and Generating Functions [3]

Partial fraction decomposition of  $G(x)$  then produces exactly Binet's formula:

$$G(x) = \frac{x}{1-x-x^2} = \frac{1}{\sqrt{5}} \left( \frac{1}{1-\varphi x} - \frac{1}{1-\psi x} \right) = \frac{1}{\sqrt{5}} \sum_{n=0}^{\infty} (\varphi^n - \psi^n) x^n$$

### Why two methods?

- **Characteristic equations** are mechanical and fast for constant-coefficient recurrences.
- **Generating functions** reveal combinatorial structure, handle variable coefficients, produce closed forms via partial fractions – at the cost of more algebra.

# Summary: Recurrence Relations

## Solution methods

1. *Telescoping* — unwind first-order recurrences
2. *Characteristic equations* — for constant-coefficient linear recurrences
3. *Generating functions* — encode the whole sequence algebraically

## Recipe: characteristic equation

Polynomial  $\rightsquigarrow$  roots  $\rightsquigarrow$  general solution  
(distinct / repeated / complex)  $\rightsquigarrow$  fit initial conditions.

## General solution forms

- Distinct:  $\alpha_1 r_1^n + \alpha_2 r_2^n$
- Repeated:  $(\alpha_1 + \alpha_2 n) r_0^n$
- Complex:  $\rho^n (\alpha \cos n\theta + \beta \sin n\theta)$
- Non-homogeneous:  $a_n = a_n^{(h)} + a_n^{(p)}$

## Connections

- Companion matrix, eigenvalues  $\rightsquigarrow$  linear algebra
- Same structure as linear differential equations
- Characteristic roots = reciprocals of GF poles

# **Annihilators**

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# Operators

**Definition 30:** An *operator* takes a function and produces another function.

The three operators we need are:

- *Sum*:  $(f + g)(n) := f(n) + g(n)$
- *Scale*:  $(\alpha \cdot f)(n) := \alpha \cdot f(n)$
- *Shift*:  $(\mathbf{E} f)(n) := f(n + 1)$

*Examples:*

- *Linearity*:  $\mathbf{E}(f + g) = \mathbf{E} f + \mathbf{E} g$ , and  $\mathbf{E}(3f) = 3 \cdot \mathbf{E} f$ .
- *Composition*:  $(\mathbf{E} - 2)f := \mathbf{E} f - 2f$ .
- *Powers*:  $\mathbf{E}^2 f(n) = f(n + 2)$ ,  $\mathbf{E}^k f(n) = f(n + k)$ .
- *Products*:  $(\mathbf{E} - 2)^2 = \mathbf{E}^2 - 4\mathbf{E} + 4$ ,  $(\mathbf{E} - 1)(\mathbf{E} - 2) = \mathbf{E}^2 - 3\mathbf{E} + 2$ .

## Applying Operators

*Examples:* Below are the results of applying different operators to  $f(n) = 2^n$ :

$$2f(n) = 2 \cdot 2^n = 2^{n+1}$$

$$3f(n) = 3 \cdot 2^n$$

$$\mathbf{E} f(n) = 2^{n+1}$$

$$\mathbf{E}^2 f(n) = 2^{n+2}$$

$$(\mathbf{E} - 2)f(n) = \mathbf{E} f(n) - 2f(n) = 2^{n+1} - 2^{n+1} = 0$$

$$(\mathbf{E}^2 - 1)f(n) = \mathbf{E}^2 f(n) - f(n) = 2^{n+2} - 2^n = 3 \cdot 2^n$$

## Compound Operators

The compound operators can be seen as polynomials in “variable”  $\mathbf{E}$ .

*Example:* The compound operators  $\mathbf{E}^2 - 3\mathbf{E} + 2$  and  $(\mathbf{E} - 1)(\mathbf{E} - 2)$  are equivalent:

$$\text{Let } g(n) := (\mathbf{E} - 2)f(n) = f(n + 1) - 2f(n)$$

$$\begin{aligned}\text{Then } (\mathbf{E} - 1)(\mathbf{E} - 2)f(n) &= (\mathbf{E} - 1)g(n) \\ &= g(n + 1) - g(n) \\ &= [f(n + 2) - 2f(n + 1)] - [f(n + 1) - 2f(n)] \\ &= f(n + 2) - 3f(n + 1) + 2f(n) \\ &= (\mathbf{E}^2 - 3\mathbf{E} + 2)f(n) \quad \checkmark\end{aligned}$$

## Operators Summary

| Operator       | Definition                                                                       |
|----------------|----------------------------------------------------------------------------------|
| addition       | $(f + g)(n) := f(n) + g(n)$                                                      |
| subtraction    | $(f - g)(n) := f(n) - g(n)$                                                      |
| multiplication | $(\alpha \cdot f)(n) := \alpha \cdot f(n)$                                       |
| shift          | $\mathbf{E} f(n) := f(n + 1)$                                                    |
| k-fold shift   | $\mathbf{E}^k f(n) := f(n + k)$                                                  |
| composition    | $(\mathbf{X} + \mathbf{Y})f := \mathbf{X} f + \mathbf{Y} f$                      |
|                | $(\mathbf{X} - \mathbf{Y})f := \mathbf{X} f - \mathbf{Y} f$                      |
|                | $\mathbf{X} \mathbf{Y} f := \mathbf{X}(\mathbf{Y} f) = \mathbf{Y}(\mathbf{X} f)$ |
| distribution   | $\mathbf{X}(f + g) = \mathbf{X} f + \mathbf{X} g$                                |

## Annihilators

**Definition 31:** An *annihilator* of a function  $f$  is any non-trivial operator  $X$  such that  $Xf = 0$ .

**Intuition:** An annihilator “kills” the function — after applying the operator, nothing remains. This is the same idea as finding roots of a polynomial: if  $(x - 2)$  is a factor, then plugging in  $x = 2$  makes the polynomial zero. Here,  $(\mathbf{E} - a)$  is a factor, and applying it to  $a^n$  makes the result zero.

*Examples:*

- $(\mathbf{E} - 1)$  annihilates any constant  $f(n) = c$ :  $(\mathbf{E} - 1)c = c - c = 0$ .
- $(\mathbf{E} - 2)$  annihilates  $f(n) = \alpha \cdot 2^n$ :  $(\mathbf{E} - 2)(\alpha 2^n) = \alpha 2^{n+1} - 2\alpha 2^n = 0$ .
- $(\mathbf{E} - 1)^2$  annihilates any linear  $f(n) = \alpha n + \beta$ :  $(\mathbf{E} - 1)(\alpha n + \beta) = \alpha$  (a constant), and  $(\mathbf{E} - 1)\alpha = 0$ .

## Annihilators Summary

| Operator                                                        | Functions annihilated                                  |
|-----------------------------------------------------------------|--------------------------------------------------------|
| $\mathbf{E} - 1$                                                | $\alpha$                                               |
| $\mathbf{E} - a$                                                | $\alpha a^n$                                           |
| $(\mathbf{E} - a)(\mathbf{E} - b)$                              | $\alpha a^n + \beta b^n$ [if $a \neq b$ ]              |
| $(\mathbf{E} - a_0)(\mathbf{E} - a_1) \dots (\mathbf{E} - a_k)$ | $\sum_{i=0}^k \alpha_i a_i^n$ [if $a_i$ are distinct]  |
| $(\mathbf{E} - 1)^2$                                            | $\alpha n + \beta$                                     |
| $(\mathbf{E} - a)^2$                                            | $(\alpha n + \beta) a^n$                               |
| $(\mathbf{E} - a)^2(\mathbf{E} - b)$                            | $(\alpha n + \beta) a^n + \gamma b^n$ [if $a \neq b$ ] |
| $(\mathbf{E} - a)^d$                                            | $(\sum_{i=0}^{d-1} \alpha_i n^i) a^n$                  |

## Properties of Annihilators

**Theorem 25:** If  $X$  annihilates  $f$ , then  $X$  also annihilates  $\alpha f$  for any constant  $\alpha$ .

**Proof:** Since operators are linear (they distribute over scaling),  $X(\alpha f) = \alpha \cdot X f = \alpha \cdot 0 = 0$ . □

**Theorem 26:** If  $X$  annihilates both  $f$  and  $g$ , then  $X$  also annihilates  $f \pm g$ .

**Proof:** By linearity of operators:  $X(f \pm g) = X f \pm X g = 0 \pm 0 = 0$ . □

**Theorem 27:** If  $X$  annihilates  $f$ , then  $X$  also annihilates  $\mathbf{E} f$ .

**Proof:** Any operator  $X$  that is a polynomial in  $\mathbf{E}$  *commutes* with  $\mathbf{E}$  itself.

Therefore  $X(\mathbf{E} f) = \mathbf{E}(X f) = \mathbf{E}(0) = 0$ . □

## Properties of Annihilators [2]

**Theorem 28:** If  $X$  annihilates  $f$  and  $Y$  annihilates  $g$ , then  $XY$  annihilates  $f \pm g$ .

**Proof:** Since  $X$  annihilates  $f$ , and by the previous theorem  $X$  also annihilates  $E f$ ,  $E^2 f$ , *etc.*, the operator  $X$  annihilates *any* function obtained by applying shift-operators to  $f$ . In particular,  $X$  annihilates  $Y f$ .

Now apply  $XY$  to  $f \pm g$ :

$$XY(f \pm g) = X(Y f \pm Y g) = X(Y f) \pm X(Y g)$$

The first term  $X(Y f)$ : since  $X$  annihilates  $f$  and commutes with  $Y$ , this is  $Y(X f) = Y(0) = 0$ . The second term  $X(Y g)$ : since  $Y$  annihilates  $g$ , we get  $X(0) = 0$ . Both terms are zero, so the result is zero.  $\square$

# Annihilating Recurrences

The annihilator method is a systematic five-step procedure for solving recurrences.



1. *Rewrite* the recurrence in operator form (replace  $a_n$  with  $\mathbf{E}$ -operators).
2. *Identify* the annihilator of both sides and multiply additional factors to kill the residue.
3. *Factor* the annihilator into linear factors ( $\mathbf{E} - a$ ).
4. *Read off* the generic solution from the factorization (one term per factor, using the summary table).
5. *Fit* the initial conditions to determine the free constants.

*Example:*  $r(n) = 5r(n-1)$  with  $r(0) = 3$ .

Step 1.  $r(n+1) - 5r(n) = 0$   
 $(\mathbf{E} - 5)r(n) = 0$

Step 2.  $(\mathbf{E} - 5)$  annihilates  $r(n)$ .

Thus,  $r(n) = 3 \cdot 5^n$ .

Step 3.  $(\mathbf{E} - 5)$  is already factored.

Step 4.  $r(n) = \alpha 5^n$  is a generic solution.

Step 5.  $r(0) = \alpha = 3 \implies \alpha = 3$

## Annihilating Recurrences [2]

*Example:*  $T(n) = 2T(n-1) + 1$  with  $T(0) = 0$

1.  $T(n+1) - 2T(n) = 1$   
 $(\mathbf{E}-2)T(n) = 1$
2.  $(\mathbf{E}-2)$  does *not* annihilate  $T(n)$ : the residue is 1.  
 $(\mathbf{E}-1)$  annihilates the residue 1.  
Thus,  $(\mathbf{E}-1)(\mathbf{E}-2)$  annihilates  $T(n)$ .
3.  $(\mathbf{E}-1)(\mathbf{E}-2)$  is already factored.
4.  $T(n) = \alpha 2^n + \beta$  is a generic solution.
5. Find the coefficients  $\alpha, \beta$  using  $T(0) = 0$  and  $T(1) = 2T(0) + 1 = 1$ :

$$\begin{cases} T(0) = 0 = \alpha \cdot 2^0 + \beta \\ T(1) = 1 = \alpha \cdot 2^1 + \beta \end{cases} \implies \begin{cases} \alpha = 1 \\ \beta = -1 \end{cases}$$

Thus,  $T(n) = 2^n - 1$ .

## Annihilating Recurrences [3]

*Example:*  $T(n) = T(n-1) + 2T(n-2) + 2^n - n^2$

1. Operator form:

$$(\mathbf{E}^2 - \mathbf{E} - 2)T(n) = \mathbf{E}^2(2^n - n^2)$$

2. Annihilator:

$$(\mathbf{E}^2 - \mathbf{E} - 2)(\mathbf{E} - 2)(\mathbf{E} - 1)^3$$

3. Factorization:

$$(\mathbf{E} + 1)(\mathbf{E} - 2)^2(\mathbf{E} - 1)^3$$

4. Generic solution:

$$T(n) = \alpha(-1)^n + (\beta n + \gamma)2^n + \delta n^2 + \varepsilon n + \zeta$$

5. There are no initial conditions. We can only provide an asymptotic bound.

Thus,  $T(n) \in \Theta(n2^n)$

## Annihilators and Characteristic Equations

**Note:** The characteristic equation method and the annihilator method are *the same algebra* in different notation. Replacing  $r$  with  $\mathbf{E}$  in the characteristic polynomial gives the annihilator, and vice versa. Understanding both gives you flexibility: use whichever is more convenient.

**Theorem 29:** For the linear homogeneous recurrence  $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$ :

- **Characteristic polynomial:**  $p(r) = r^k - c_1 r^{k-1} - \dots - c_k$
- **Annihilator polynomial:**  $p(\mathbf{E}) = \mathbf{E}^k - c_1 \mathbf{E}^{k-1} - \dots - c_k$

If  $r_1, \dots, r_t$  are the distinct characteristic roots with multiplicities  $m_1, \dots, m_t$ , then

$$p(r) = (r - r_1)^{m_1} \dots (r - r_t)^{m_t} = p(\mathbf{E})$$

The factorization is identical; only the interpretation differs.

**Proof:** The recurrence  $a_n - c_1 a_{n-1} - \dots - c_k a_{n-k} = 0$  can be written as  $(\mathbf{E}^k - c_1 \mathbf{E}^{k-1} - \dots - c_k) a_n = 0$ . This is the *annihilator polynomial* applied to  $a_n$ .

## Annihilators and Characteristic Equations [2]

The characteristic equation is  $r^k - c_1 r^{k-1} - \dots - c_k = 0$ .

Both polynomials have the same coefficients — the only difference is the symbol ( $r$  vs.  $\mathbf{E}$ ). Therefore they have the same roots and the same factorization. □

*Example:*  $a_n = 5a_{n-1} - 6a_{n-2}$ .

- **Characteristic:**  $r^2 - 5r + 6 = (r - 2)(r - 3) = 0$ .
- **Annihilator:**  $(\mathbf{E} - 2)(\mathbf{E} - 3)a_n = 0$ .
- Both give:  $a_n = \alpha 2^n + \beta 3^n$ .

## Common Mistakes

*Example:* Solve  $a_n = 2a_{n-1} - 2^n$  with  $a_0 = 1$ .

**Wrong answer:** “Characteristic equation:  $r - 2 = 0$ , so  $r = 2$ . General solution:  $a_n = c \cdot 2^n$ .”

**Where's the error?**

- This is *non-homogeneous* (has extra term “ $-2^n$ ”), but the student *ignored it!*
- We need a *particular* solution in addition to the homogeneous one.
- Since 2 is *also* the characteristic root (resonance!), the simple trial  $\beta 2^n$  fails. We must try  $a_n^{(p)} = \beta n \cdot 2^n$ .

**Correct:**  $a_n = (\alpha + n)2^n$ . Using  $a_0 = 1$ :  $\alpha = 1$ , so  $a_n = (1 + n)2^n$ .

## Summary: Annihilators

**Core idea.** An operator  $X$  *annihilates*  $f$  if  $X f = 0$ .

The factor  $(\mathbf{E} - a)$  kills  $a^n$ . Higher powers  $(\mathbf{E} - a)^d$  kill  $(\alpha_0 + \alpha_1 n + \dots + \alpha_{d-1} n^{d-1}) a^n$ .

Finding the annihilator of a recurrence *automatically* reveals the form of the solution.

### Algorithm

1. Rewrite recurrence in operator form
2. Find the annihilator of both sides
3. Factor into  $(\mathbf{E} - a_i)$  terms
4. Read off generic solution from factors
5. Fit initial conditions

### Key properties

- If  $X$  annihilates  $f$ ,  
then it also annihilates  $\alpha f, f \pm g, \mathbf{E} f$
- If  $X$  annihilates  $f$  and  $Y$  annihilates  $g$ ,  
then  $X Y$  annihilates  $f \pm g$
- Same polynomial as characteristic equation, with  
 $r \Rightarrow \mathbf{E}$

# Asymptotic Analysis

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*“A mathematician who is not also something of a poet  
will never be a complete mathematician.”*

*— Karl Weierstrass*

# Asymptotic Notation

**Definition 32** (*Big-O notation*):  $f(n) \in O(g(n))$  means  $f$  is *asymptotically bounded from above* by  $g$ , up to a constant factor:

$$f(n) \in O(g(n)) \iff \exists c > 0. \exists n_0. \forall n > n_0 : |f(n)| \leq c \cdot g(n)$$

**Definition 33** (*Small-o notation*):  $f(n) \in o(g(n))$  means  $f$  is *strictly dominated* by  $g$ : no constant makes  $f$  keep up with  $g$ .

$$f(n) \in o(g(n)) \iff \forall c > 0. \exists n_0. \forall n > n_0 : |f(n)| \leq c \cdot g(n)$$

**Note:** The only difference is  $\exists c$  (Big-O: *some* constant works) vs.  $\forall c$  (Small-o: *every* constant works, no matter how small).

**Note:** Flip  $\leq$  to  $\geq$  to get the dual notations  $f \in \Omega(g)$  (bounded below) and  $f \in \omega(g)$  (strictly dominates).

## Asymptotic Notation [2]

**Definition 34** (*Theta notation*):  $f \in \Theta(g)$  iff  $f \in O(g)$  and  $f \in \Omega(g)$  –  $f$  and  $g$  grow at the *same rate*, up to constant factors.

*Example:* Prove that  $3n^2 + 5n \in \Theta(n^2)$ .

**Upper bound:**  $3n^2 + 5n \leq 3n^2 + 5n^2 = 8n^2$  for  $n \geq 1$ , so  $c = 8$ ,  $n_0 = 1$ .

**Lower bound:**  $3n^2 + 5n \geq 3n^2$  for  $n \geq 0$ , so  $c = 3$ ,  $n_0 = 0$ .

Both bounds hold, so  $3n^2 + 5n \in \Theta(n^2)$ .

## Limits

| Notation          | Name        | Description                        | Limit definition                                              |
|-------------------|-------------|------------------------------------|---------------------------------------------------------------|
| $f \in o(g)$      | Small Oh    | $f$ is dominated by $g$            | $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0$           |
| $f \in O(g)$      | Big Oh      | $f$ is bounded above by $g$        | $\limsup_{n \rightarrow \infty} \frac{ f(n) }{g(n)} < \infty$ |
| $f \sim g$        | Equivalence | $f$ is asymptotically equal to $g$ | $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$           |
| $f \in \Omega(g)$ | Big Omega   | $f$ is bounded below by $g$        | $\liminf_{n \rightarrow \infty} \frac{f(n)}{g(n)} > 0$        |
| $f \in \omega(g)$ | Small Omega | $f$ dominates $g$                  | $\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$      |

## Asymptotic Equivalence

**Definition 35:**  $f \sim g$  means  $f$  and  $g$  are *asymptotically equivalent* — their ratio tends to 1:

$$f \sim g \iff \lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 1$$

*Example:*  $n^2 + 3n \sim n^2$ , since  $\frac{n^2+3n}{n^2} = 1 + \frac{3}{n} \rightarrow 1$ .

**Note:**  $f \sim g$  is *stronger* than  $f \in \Theta(g)$ :  $2n \in \Theta(n)$  but  $2n \sim n$  is false (the ratio is 2, not 1).

## Properties of Asymptotics

**Theorem 30:** The asymptotic notations satisfy:

$$f \in O(g) \text{ and } f \in \Omega(g) \iff f \in \Theta(g)$$

$$f \in O(g) \iff g \in \Omega(f)$$

$$f \in o(g) \iff g \in \omega(f)$$

$$f \in o(g) \rightarrow f \in O(g)$$

$$f \in \omega(g) \rightarrow f \in \Omega(g)$$

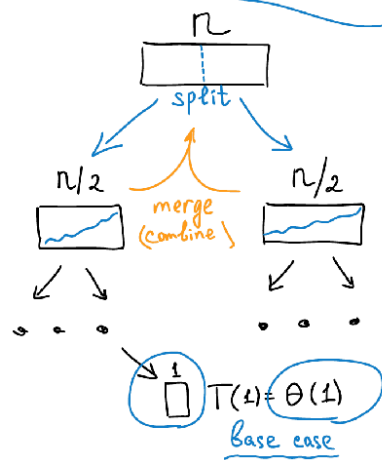
$$f \sim g \rightarrow f \in \Theta(g)$$

**Note: Growth rate hierarchy:**  $1 \subset o(\ln n) \subset o(n) \subset o(n \ln n) \subset o(n^2) \subset o(2^n) \subset o(n!)$ .

Each class is strictly contained in the next:  $n^k \in o(c^n)$  for any  $k$  and  $c > 1$ .

## Divide-and-Conquer Algorithms Analysis

Divide-and-Conquer



$$T(n) = \underbrace{2 T(n/2)}_{\text{recursive work}} + \underbrace{\Theta(n)}_{\text{split/merge work}}$$

$$\left\{ \begin{aligned} \Sigma &= 2 \cdot 2 \cdot \underbrace{T(n/4)}_{\text{merge}} = n \end{aligned} \right.$$

$$\left\{ \begin{aligned} \Sigma &\approx 2^{i+1} T(n/2^{i+1}) = n \end{aligned} \right.$$

$$\left\{ \begin{aligned} \Sigma &= n \cdot \Theta(1) = n \end{aligned} \right.$$

$$\underbrace{\Theta(n)}_{\text{work on each level of recursion tree}} \cdot \underbrace{\Theta(\log n)}_{\text{height of recursion tree}} = \Theta(n \log n)$$

## Divide-and-Conquer Recurrence

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

- $T(n)$  is the *cost* of the recursive algorithm
- $a$  is the number of *parts* (*sub-problems*)
- $n/b$  is the *size* of each part
- $T\left(\frac{n}{b}\right)$  is the cost of each *sub-problem*
- $f(n)$  is the cost of *splitting* and *merging* the solutions of the subproblems

Hereinafter,  $c_{\text{crit}} = \log_b a$  is a *critical constant*.

## Master Theorem

The master theorem [2] gives a direct asymptotic bound for recurrences of the form

$$T(n) = a \cdot T\left(\frac{n}{b}\right) + f(n)$$

where  $c_{\text{crit}} = \log_b a$  is the *critical exponent*.

**Intuition:** The recurrence splits into  $a$  sub-problems of size  $n/b$ . The total work at each level of the recursion tree is  $a$  copies of the sub-problem cost, plus the merge cost  $f(n)$ .

- If  $f(n)$  is small relative to  $n^{c_{\text{crit}}}$ , the recursion *leaves* dominate — total cost is  $\Theta(n^{c_{\text{crit}}})$ .
- If  $f(n)$  is comparable to  $n^{c_{\text{crit}}}$ , the levels contribute equally — a log factor appears.
- If  $f(n)$  is large, the *root level* dominates — total cost is  $\Theta(f(n))$ .

## Master Theorem [2]

| Case     | Description                   | Condition                                                 | Bound                                               |
|----------|-------------------------------|-----------------------------------------------------------|-----------------------------------------------------|
| Case I   | “merge” $\ll$ “recursion”     | $f(n) \in O(n^c), c < c_{\text{crit}}$                    | $T(n) \in \Theta(n^{c_{\text{crit}}})$              |
| Case II  | “merge” $\approx$ “recursion” | $f(n) \in \Theta(n^{c_{\text{crit}}} \log^k n), k \geq 0$ | $T(n) \in \Theta(n^{c_{\text{crit}}} \log^{k+1} n)$ |
| Case III | “merge” $\gg$ “recursion”     | $f(n) \in \Omega(n^c), c > c_{\text{crit}}$               | $T(n) \in \Theta(f(n))$                             |

**Note:** Case III also requires the *regularity condition* to hold:  $af(n/b) \leq kf(n)$  for some constant  $k < 1$  and all sufficiently large  $n$ .

## Examples of Master Theorem Application

*Example:*  $T(n) = 2T(n/4) + n^{0.51}$

$a = 2, b = 4, c_{\text{crit}} = \log_4 2 = \frac{1}{2}$ .  $f(n) = n^{0.51} \in \Omega(n^c)$  with  $c = 0.51 > 0.5 = c_{\text{crit}}$ .

Case III:  $T(n) \in \Theta(n^{0.51})$ .

*Example:*  $T(n) = 5T(n/25) + n^{0.49}$

$a = 5, b = 25, c_{\text{crit}} = \log_{25} 5 = \frac{1}{2}$ .  $f(n) = n^{0.49} \in O(n^c)$  with  $c = 0.49 < 0.5 = c_{\text{crit}}$ .

Case I:  $T(n) \in \Theta(n^{\frac{1}{2}})$ .

*Example:*  $T(n) = 3T(n/9) + \sqrt{n}$

$a = 3, b = 9, c_{\text{crit}} = \log_9 3 = \frac{1}{2}$ .  $f(n) = n^{\frac{1}{2}} = n^{c_{\text{crit}}}$ , so  $f(n) \in \Theta(n^{c_{\text{crit}}} \log^0 n)$ .

Case II ( $k = 0$ ):  $T(n) \in \Theta(n^{\frac{1}{2}} \log n)$ .

## Akra–Bazzi Method

The Akra–Bazzi method [3] is a *generalization* of the master theorem to recurrences of the form

$$T(n) = f(n) + \sum_{i=1}^k a_i T\left(b_i n + \underbrace{h_i(n)}_{*}\right)$$

- $k$  is a constant
- $a_i > 0$
- $0 < b_i < 1$
- $h_i(n) \in O\left(\frac{n}{\log^2 n}\right)$  is a *small perturbation*

Bound of  $T(n)$  by Akra–Bazzi method:

$$T(n) \in \Theta\left(n^p \cdot \left(1 + \int_1^n \frac{f(x)}{x^{p+1}} dx\right)\right)$$

where  $p$  is the solution for the equation  $\sum_{i=1}^k a_i b_i^p = 1$

## Akra–Bazzi Example

*Example:* Suppose the runtime of an algorithm is expressed by the following recurrence relation:

$$T(n) = \begin{cases} 1 & \text{for } 0 \leq n \leq 3 \\ n^2 + \frac{7}{4}T(\lfloor \frac{1}{2}n \rfloor) + T(\lceil \frac{3}{4}n \rceil) & \text{for } n > 3 \end{cases}$$

- Note that the Master Theorem *is not* applicable here, since there are *two* different recursive terms.
- Let's apply the Akra–Bazzi method. First, solve the equation  $\frac{7}{4}(\frac{1}{2})^p + (\frac{3}{4})^p = 1$ . This gives us  $p = 2$ .
- Next, use the formula from AB-method to obtain the bound:

$$\begin{aligned} T(x) &\in \Theta\left(x^p \left(1 + \int_1^x \frac{f(u)}{u^{p+1}} du\right)\right) = \\ &= \Theta\left(x^2 \left(1 + \int_1^x \frac{u^2}{u^3} du\right)\right) = \\ &= \Theta(x^2(1 + \ln x)) = \\ &= \Theta(x^2 \log x) \end{aligned}$$

# Summary: Asymptotic Analysis

## Notations

- $O(g)$ : bounded above by  $g$  (some constant  $c$ )
- $\Omega(g)$ : bounded below by  $g$
- $\Theta(g)$ : bounded above *and* below — tight
- $o(g)$ : strictly dominated (no constant saves you)
- $\omega(g)$ : strictly dominates
- $f \sim g$ : ratio  $\rightarrow 1$  (stronger than  $\Theta$ )

**Growth hierarchy:**  $1 \subset o(\ln n) \subset o(n) \subset o(n \ln n) \subset o(n^2) \subset o(2^n) \subset o(n!)$

## Master Theorem

$T(n) = aT(n/b) + f(n)$ , where  $c_{\text{crit}} = \log_b a$ .

Compare  $f(n)$  with  $n^{c_{\text{crit}}}$ :

- Case I:  $f$  is smaller  $\rightarrow \Theta(n^{c_{\text{crit}}})$
- Case II:  $f$  is comparable  $\rightarrow \Theta(n^{c_{\text{crit}}} \log n)$
- Case III:  $f$  dominates  $\rightarrow \Theta(f(n))$

## Akra-Bazzi

Generalizes Master Theorem to unequal subproblem sizes. Solve  $\sum a_i b_i^p = 1$  for  $p$ , then integrate.

# **Advanced Topics**

## Why Extend the Factorial?

The factorial  $n! = 1 \cdot 2 \cdot \dots \cdot n$  is defined only for non-negative integers. But many formulas *need* factorial-like values at non-integer points:

- **Generalized binomial coefficients:**  $\binom{r}{k}$  for real  $r$  requires a continuous version of  $r!$ .
- **Probability:** the Gaussian integral  $\int e^{-t^2} dt$  evaluates to a non-integer “factorial” value.
- **Asymptotics:** Stirling’s formula  $n! \approx \sqrt{2\pi n}(n/e)^n$  comes from the analytic properties of  $\Gamma$ .

How should we *interpolate*  $n!$  between the integers? There are infinitely many smooth extensions; we need a principled way to choose.

## Euler Integral Definition

**Definition 36:** The *Gamma function* is defined for  $z \in \mathbb{C}$  with  $\Re(z) > 0$  by the *Euler integral*:

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

The integral converges for all  $z$  with  $\Re(z) > 0$  because the exponential  $e^{-t}$  decays faster than any power of  $t$  grows.

*Example:*  $\Gamma(1) = \int_0^{\infty} e^{-t} dt = 1$ . A change of variable  $t = s^2$  gives  $\Gamma(1/2) = \int_0^{\infty} e^{-s^2} ds = \sqrt{\pi}$ .

## The Functional Equation

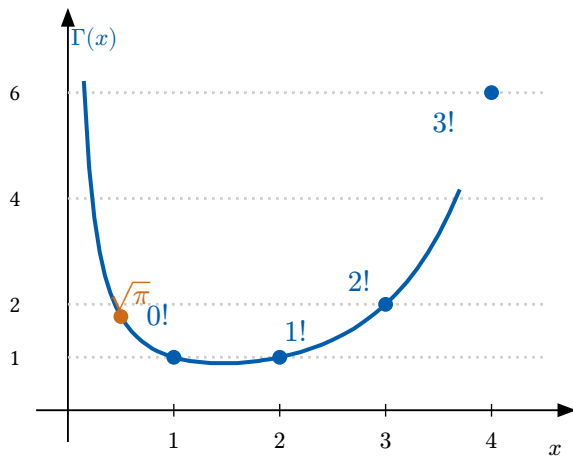
**Theorem 31:** For all  $z$  with  $\Re(z) > 0$ :  $\Gamma(z+1) = z \cdot \Gamma(z)$ . In particular,  $\Gamma(n+1) = n!$  for every positive integer  $n$ .

**Proof:** Start from  $\Gamma(z+1) = \int_0^\infty t^z e^{-t} dt$ . Integrate by parts with  $u = t^z$ ,  $dv = e^{-t} dt$ :

$$\Gamma(z+1) = [-t^z e^{-t}]_0^\infty + z \int_0^\infty t^{z-1} e^{-t} dt$$

The boundary term vanishes: at  $t = 0$  we get  $t^z e^{-t} = 0$  (since  $z > 0$ ), and at  $t = \infty$  the exponential decay dominates. Therefore  $\Gamma(z+1) = z \cdot \Gamma(z)$ . Iterating gives  $\Gamma(n+1) = n \cdot \dots \cdot 1 \cdot \Gamma(1) = n!$ , since  $\Gamma(1) = \int_0^\infty e^{-t} dt = 1$ . □

## Plot of $\Gamma(x)$



### Key values

1.  $\Gamma(1) = 1 = 0!$
2.  $\Gamma(2) = 1 = 1!$
3.  $\Gamma(3) = 2 = 2!$
4.  $\Gamma(4) = 6 = 3!$
5.  $\Gamma(1/2) = \sqrt{\pi} \approx 1.77$

### Behavior

- Pole at  $x = 0^+$ :  $\Gamma(x) \rightarrow +\infty$
- Minimum near  $x \approx 1.46$ :  $\Gamma \approx 0.89$
- Poles at all  $x = 0, -1, -2, \dots$

## Bohr–Mollerup Theorem

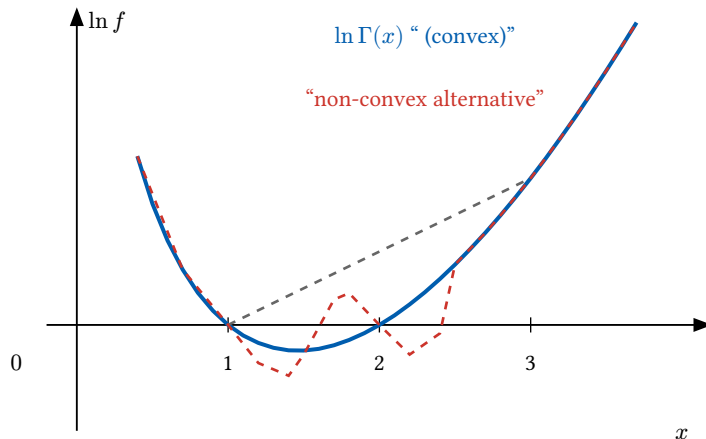
**Theorem 32:** There is a *unique* function  $f : (0, \infty) \rightarrow (0, \infty)$  satisfying all three conditions:

1.  $f(1) = 1$
2.  $f(x + 1) = x \cdot f(x)$  for all  $x > 0$
3.  $\ln f(x)$  is convex

That unique function is  $f(x) = \Gamma(x)$ .

The first two conditions alone are not enough: there are infinitely many extensions of  $n!$  satisfying the functional equation. Convexity of  $\ln \Gamma$  is the natural smoothness requirement, ruling out functions that wiggle wildly between integer points. By Bohr–Mollerup, any function satisfying these three conditions *must be*  $\Gamma$ . This will guarantee that the Gauss limit and Weierstrass product (introduced next) define the same function as the Euler integral.

## Convexity of $\ln \Gamma$



The chord (gray dashed) lies *above*  $\ln \Gamma$ , confirming convexity. The red dashed curve is an alternative extension that satisfies the first two Bohr–Mollerup conditions but is *not* convex.

## Gauss's Limit Form

**Definition 37:** Gauss's *limit form* defines  $\Gamma(x)$  for  $x > 0$  as:

$$\Gamma(x) := \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{\prod_{k=0}^n (x + k)}$$

**Theorem 33:** The Gauss limit satisfies the same functional equation:  $\Gamma(x + 1) = x \cdot \Gamma(x)$ .

**Proof:** Write  $\Gamma(x + 1)$  using the limit:

$$\Gamma(x + 1) = \lim_{n \rightarrow \infty} \frac{n! \cdot n^{x+1}}{\prod_{k=0}^n (x + 1 + k)}$$

Shift the index:  $\prod_{k=0}^n (x + 1 + k) = \prod_{k=1}^{n+1} (x + k)$ . Factor out  $\Gamma(x)$  and the extra terms:

## Gauss's Limit Form [2]

$$= \lim_{n \rightarrow \infty} x \cdot \frac{n! \cdot n^x}{\prod_{k=0}^n (x+k)} \cdot \frac{n}{x+n+1}$$

As  $n \rightarrow \infty$ , the factor  $n/(x+n+1) \rightarrow 1$ , and the remaining limit is  $x \cdot \Gamma(x)$  by definition. □

## Weierstrass Product

**Definition 38:** The *Weierstrass product* defines  $1/\Gamma(x)$  as an infinite product:

$$\frac{1}{\Gamma(x)} = x \cdot e^{\gamma x} \cdot \prod_{n=1}^{\infty} \left( \left(1 + \frac{x}{n}\right) e^{-x/n} \right)$$

where  $\gamma = \lim_{n \rightarrow \infty} \left( \sum_{k=1}^n 1/k - \ln n \right) \approx 0.5772$  is the *Euler–Mascheroni constant*.

This form defines  $1/\Gamma(x)$  as an entire function (no poles), so the poles of  $\Gamma(x)$  at  $x = 0, -1, -2, \dots$  come from the simple zeros of the product. Since all three definitions satisfy the Bohr–Mollerup conditions, they define the same function.

## Equivalence of Definitions

**Theorem 34:** The Euler integral and Gauss limit definitions agree: for all  $x > 0$ ,

$$\int_0^\infty t^{x-1} e^{-t} dt = \lim_{n \rightarrow \infty} \frac{n! \cdot n^x}{\prod_{k=0}^n (x+k)}$$

**Proof:** Define  $I_n := \int_0^n t^{x-1} (1 - t/n)^n dt$ . As  $n \rightarrow \infty$ ,  $(1 - t/n)^n \rightarrow e^{-t}$ , so  $I_n \rightarrow \Gamma(x)$ .

Substitute  $u = t/n$ :

$$I_n = n^x \int_0^1 u^{x-1} (1 - u)^n du = n^x \cdot B(x, n+1)$$

where  $B(x, y) = \int_0^1 t^{x-1} (1 - t)^{y-1} dt$ .

## Equivalence of Definitions [2]

Using the Beta–Gamma relation  $B(x, y) = (\Gamma(x) \cdot \Gamma(y)) / \Gamma(x + y)$  (itself provable from the Euler integral):

$$I_n = n^x \cdot \frac{\Gamma(x) \cdot \Gamma(n + 1)}{\Gamma(x + n + 1)} = \frac{n! \cdot n^x}{x \cdot (x + 1) \cdot \cdots (x + n)}$$

since  $\Gamma(n + 1) = n!$  and  $\Gamma(x + n + 1) = (x + n)(x + n - 1) \cdots x \cdot \Gamma(x)$ .

Taking  $n \rightarrow \infty$  and matching both sides completes the proof. □

## Euler's Reflection Formula

**Theorem 35:** For all non-integer  $x$ :

$$\Gamma(x) \cdot \Gamma(1-x) = \frac{\pi}{\sin(\pi x)}$$

*Example:*  $\Gamma(1/2) \cdot \Gamma(1/2) = \frac{\pi}{\sin(\pi/2)} = \pi$ , so  $\Gamma(1/2) = \sqrt{\pi}$ . This confirms that the factorial extends to half-integers:  $(1/2)! = \Gamma(3/2) = \sqrt{\pi}/2$ .

## Stirling's Approximation

**Theorem 36:** As  $n \rightarrow \infty$ :

$$n! \sim \sqrt{2\pi n}(n/e)^n$$

More precisely,  $n! = \sqrt{2\pi n}(n/e)^n e^{\theta_n}$  where  $1/(12n+1) < \theta_n < 1/(12n)$ .

Stirling's approximation lets us estimate factorials and binomial coefficients without computing huge products.

For example,  $\binom{2n}{n} \approx 4^n / \sqrt{\pi n}$  follows directly.

*Example:* Estimate  $10!$  using Stirling:  $\sqrt{20\pi}(10/e)^{10} \approx 3.599 \times 10^6$ . The exact value is  $10! = 3628800 \approx 3.629 \times 10^6$ , error less than 1%.

## Gamma Function Applications

The Gamma function appears throughout combinatorics and analysis.

- **Generalized binomial coefficients** (for arbitrary  $r \in \mathbb{R}$ ):  $\binom{r}{k} = \frac{\Gamma(r+1)}{\Gamma(k+1) \cdot \Gamma(r-k+1)}$ , the definition underlying Newton's Binomial Theorem for real exponents.
- **Beta function:**  $B(x, y) = \frac{\Gamma(x)\Gamma(y)}{\Gamma(x+y)} = \int_0^1 t^{x-1}(1-t)^{y-1} dt$ .
- **Volume of the  $n$ -ball:**  $V_n = \pi^{n/2} / \Gamma(n/2 + 1)$ , a natural application of  $\Gamma$  at half-integer arguments.
- **Asymptotics of partitions and Stirling numbers:** leading terms involve  $\Gamma$  through saddle-point methods.

# Summary: Gamma Function

## Why $\Gamma$ is unique (Bohr-Mollerup)

1.  $\Gamma(1) = 1$
2.  $\Gamma(x+1) = x \cdot \Gamma(x)$
3.  $\ln \Gamma(x)$  is convex

## Three equivalent definitions

- Euler integral:  $\int_0^\infty t^{z-1} e^{-t} dt$
- Gauss limit:  $\lim (n! n^x) / (\prod (x+k))$
- Weierstrass:  $1/\Gamma = x e^{\gamma x} \prod ((1 + x/n) e^{-x/n})$

## Key properties

- Functional equation:  $\Gamma(x+1) = x \cdot \Gamma(x)$
- Integer values:  $\Gamma(n) = (n-1)!$
- Reflection:  $\Gamma(x)\Gamma(1-x) = \pi / \sin(\pi x)$
- $\Gamma(1/2) = \sqrt{\pi}$

## Applications

- Extends  $n!$  to reals and complexes
- Generalizes  $\binom{n}{k}$  to real  $n$
- Stirling:  $n! \approx \sqrt{2\pi n} (n/e)^n$

# Möbius Inversion

---

*“The opposite of a correct statement is a false statement.  
But the opposite of a profound truth may well be another profound truth.”*

*— Niels Bohr*



Gian-Carlo  
Rota



August  
Ferdinand  
Möbius

## Motivation

Throughout this course we have derived many counting formulas:

- Multiplication principle:  $k^n$
- Permutations:  $n!/(n-k)!$
- Combinations:  $\binom{n}{k}$
- Stars and bars:  $\binom{n+k-1}{n}$

- Stirling numbers:  $s_k^{\Pi}(n)$
- Bell numbers:  $B_n$
- PIE:  $\sum (-1)^i \binom{n}{i} (n-i)^k$
- Derangements:  $D_n$

They all seem unrelated.

But many of them are unified by *one* deep pattern: **Möbius inversion on partially ordered sets.**

# Posets

**Definition 39:** A *partially ordered set* (poset) is a pair  $\mathcal{P} = (P, \leq)$  where  $\leq$  is a reflexive, antisymmetric, and transitive binary relation.

We will work with three key examples:

## Boolean lattice

$$(2^{[n]}, \subseteq)$$

Elements: subsets of  $[n]$ .

$$A \leq B \text{ iff } A \subseteq B.$$

## Divisor lattice

$$(\mathbb{Z}^+, |)$$

Elements: positive integers.

$$a \leq b \text{ iff } a \mid b.$$

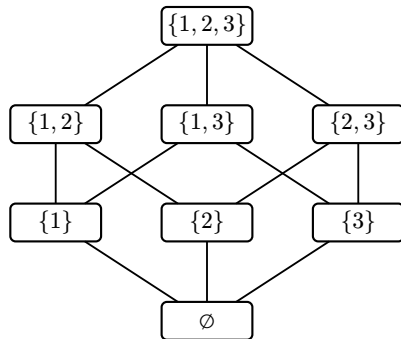
## Partition lattice

$$\Pi_n$$

Elements: partitions of  $[n]$ .

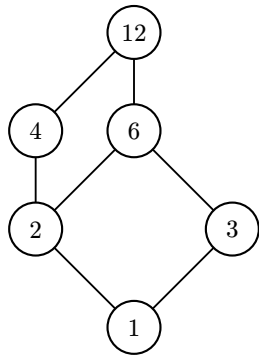
$$\pi \leq \sigma \text{ iff } \pi \text{ refines } \sigma.$$

## The Boolean Lattice $B_3$



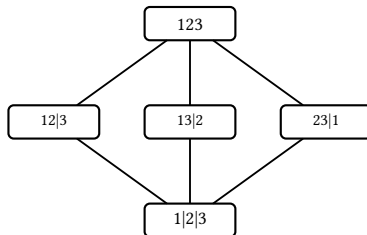
The Boolean lattice  $B_3 = (2^{[3]}, \subseteq)$ : elements are subsets of  $[3] = \{1, 2, 3\}$ , ordered by inclusion.

## The Divisor Lattice of 12



The divisor lattice of 12: elements are divisors of  $12 = 2^2 \cdot 3$ , ordered by divisibility.

## The Partition Lattice $\Pi_3$



The partition lattice  $\Pi_3$ : elements are partitions of  $[3]$ , ordered by refinement.  $\pi \leq \sigma$  means every block of  $\pi$  is contained in a block of  $\sigma$ .

Stirling number  $s_k^{\Pi}(n)$  counts the elements of  $\Pi_n$  with exactly  $k$  blocks.

## Intervals

**Definition 40:** Let  $\mathcal{P} = (P, \leq)$  be a poset and  $x \leq y$ . The *interval*  $[x, y]$  is the set:

$$[x, y] = \{z \in P \mid x \leq z \leq y\}$$

A poset is *locally finite* if every interval is finite.

*Example:* In the Boolean lattice  $(2^{[3]}, \subseteq)$ :

$$[\{1\}, \{1, 2, 3\}] = \{\{1\}, \{1, 2\}, \{1, 3\}, \{1, 2, 3\}\}$$

In the divisor lattice of 12:  $[2, 12] = \{2, 4, 6, 12\}$

## The Incidence Algebra

**Definition 41:** Let  $\mathcal{P} = (P, \leq)$  be a locally finite poset. The *incidence algebra*  $I(\mathcal{P}; R)$  consists of functions  $f : \{(x, y) \in P^2 \mid x \leq y\} \rightarrow R$  with the *convolution product*:

$$(f * g)(x, y) = \sum_{x \leq z \leq y} f(x, z) \cdot g(z, y)$$

### Matrix intuition.

1. Index elements  $x_1, x_2, \dots$  so that  $x_i \leq x_j$  implies  $i \leq j$ .
2. Each function becomes an *upper-triangular matrix*.
3. Convolution is just *matrix multiplication*.

## Three Key Functions

The incidence algebra has three elements that make everything work.

**Definition 42:** The *Kronecker delta* is the identity for convolution:

$$\delta(x, y) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{otherwise} \end{cases}$$

**Definition 43:** The *zeta function* encodes the order relation:

$$\zeta(x, y) = \begin{cases} 1 & \text{if } x \leq y \\ 0 & \text{otherwise} \end{cases}$$

**Definition 44:** The *Möbius function* is the inverse of  $\zeta$  in the incidence algebra:

$$\mu * \zeta = \zeta * \mu = \delta$$

## Three Key Functions [2]

Convolving with  $\zeta$  *sums over everything below*: for any  $f$ ,  $(f * \zeta)(y) = \sum_{x \leq y} f(x)$ .

Convolving with  $\mu$  *undoes* that sum.

### Analogy:

- $\zeta$  = “integrate” (accumulate)
- $\mu$  = “differentiate” (recover original)
- $\delta$  = “do nothing”

## Recursive Formula

The algebraic definition  $\mu * \zeta = \delta$  yields an explicit recursion:

$$\mu(x, x) = 1 \text{ for all } x \in P$$

For  $x < y$ :

$$\mu(x, y) = - \sum_{x \leq z < y} \mu(x, z)$$

*Example (Möbius function on the chain [3]):* For the chain  $(\{1, 2, 3\}, \leq)$ :

- $\mu(1, 1) = 1, \mu(2, 2) = 1, \mu(3, 3) = 1$
- $\mu(1, 2) = -\mu(1, 1) = -1$
- $\mu(2, 3) = -\mu(2, 2) = -1$
- $\mu(1, 3) = -(\mu(1, 1) + \mu(1, 2)) = -(1 + (-1)) = 0$

## Computing $\mu$ on the Boolean Lattice $B_3$

| $B$           | $ B $ | $\mu(\emptyset, B)$ | Computation                    |
|---------------|-------|---------------------|--------------------------------|
| $\emptyset$   | 0     | 1                   | base case                      |
| $\{1\}$       | 1     | -1                  | $-\mu(\emptyset, \emptyset)$   |
| $\{2\}$       | 1     | -1                  | $-\mu(\emptyset, \emptyset)$   |
| $\{3\}$       | 1     | -1                  | $-\mu(\emptyset, \emptyset)$   |
| $\{1, 2\}$    | 2     | 1                   | $-(1 + (-1) + (-1))$           |
| $\{1, 3\}$    | 2     | 1                   | $-(1 + (-1) + (-1))$           |
| $\{2, 3\}$    | 2     | 1                   | $-(1 + (-1) + (-1))$           |
| $\{1, 2, 3\}$ | 3     | -1                  | $-(1 - 1 - 1 - 1 + 1 + 1 + 1)$ |

Non-bottom intervals confirm the same pattern:

- $\mu(\{1\}, \{1, 2\}) = -1 = (-1)^1$  (distance 1)
- $\mu(\{1\}, \{1, 2, 3\}) = -(1 + (-1) + (-1)) = 1 = (-1)^2$  (distance 2)
- $\mu(\{1, 2\}, \{1, 2, 3\}) = -1 = (-1)^1$  (distance 1)

The pattern is clear:  $\mu(A, B) = (-1)^{|B|-|A|}$ .

## Möbius Inversion Formula

The Möbius function satisfies the *cancellation identity*: for any  $x \leq y$ ,

$$\sum_{x \leq z \leq y} \mu(x, z) = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x < y \end{cases}$$

**Theorem 37:** Let  $\mathcal{P} = (P, \leq)$  be a finite poset and  $f, g : P \rightarrow R$ . Then:

$$f(x) = \sum_{y \leq x} g(y) \iff g(x) = \sum_{y \leq x} \mu(y, x) \cdot f(y)$$

In words:

1.  $f$  counts “everything below”  $g$ .
2.  $\mu$  tells you how to *recover*  $g$  from  $f$ .

## Proof of the Inversion Formula

**Proof:** ( $\Rightarrow$ ) Suppose  $f(x) = \sum_{y \leq x} g(y) = (g * \zeta)(x)$ . Convolve both sides with  $\mu$ :

$$\begin{aligned} f * \mu &= (g * \zeta) * \mu \\ &= g * (\zeta * \mu) \\ &= g * \delta = g \end{aligned}$$

So  $g(x) = (f * \mu)(x) = \sum_{y \leq x} f(y) \cdot \mu(y, x)$ .

( $\Leftarrow$ ) Conversely, if  $g = f * \mu$ , then  $g * \zeta = f * \mu * \zeta = f * \delta = f$ . □

## Möbius Function on the Boolean Lattice

Consider the Boolean lattice  $\mathcal{P} = (2^{[n]}, \subseteq)$ .

**Theorem 38:** In the Boolean lattice, the Möbius function is:

$$\mu(A, B) = (-1)^{|B|-|A|}$$

for  $A \subseteq B$ .

**Proof:** By strong induction on  $d = |B| - |A|$ .

**Base** ( $d = 0$ ):  $A = B$ , so  $\mu(A, A) = 1 = (-1)^0$ . ✓

**Inductive step:** For  $A \subset B$  with  $|B| - |A| = d \geq 1$ :

$$\mu(A, B) = - \sum_{A \subsetneq C \subset B} \mu(A, C)$$

## Möbius Function on the Boolean Lattice [2]

There are  $\binom{d}{i}$  sets  $C$  with  $|C| - |A| = i$  (choose which  $i$  elements to add).

By the inductive hypothesis, each contributes  $(-1)^i$ :

$$= - \sum_{i=0}^{d-1} \binom{d}{i} (-1)^i$$

Adding the missing  $i = d$  term and subtracting it back:

$$\begin{aligned} &= - \left( \sum_{i=0}^d \binom{d}{i} (-1)^i - (-1)^d \right) \\ &= -(0 - (-1)^d) = (-1)^d \end{aligned}$$

since  $(1 - 1)^d = 0$  for  $d \geq 1$ . ✓

□

## PIE as Möbius Inversion

Plugging  $\mu(A, B) = (-1)^{|B|-|A|}$  into the inversion formula:

If  $f(S) = \sum_{T \subseteq S} g(T)$  for all  $S$ , then:

$$g(S) = \sum_{T \subseteq S} (-1)^{|S|-|T|} f(T)$$

### PIE IS Möbius inversion on the Boolean lattice!

The alternating signs  $(-1)^{|S|}$  come directly from  $\mu(A, B) = (-1)^{|B|-|A|}$ .

This is the *deep reason* PIE works: it is a special case of a general algebraic identity.

## The Chain: Finite Differences

Consider the simplest poset:  $\mathbb{N}$  with the usual order  $\leq$ .

The Möbius function here is:

$$\mu(x, y) = \begin{cases} 1 & \text{if } y = x \\ -1 & \text{if } y = x + 1 \\ 0 & \text{otherwise} \end{cases}$$

Plugging into the inversion formula:

1. If  $G(n) = \sum_{k=0}^n F(k)$  (accumulated sum),
2. then  $F(n) = G(n) - G(n - 1)$  (finite difference).

Möbius inversion on the chain is the *fundamental theorem of discrete calculus*.

Summation and differencing are inverse operations — just like integration and differentiation are in continuous calculus.

## Zhegalkin Polynomials

The *Zhegalkin polynomial* (algebraic normal form) of a Boolean function  $f : \{0, 1\}^n \rightarrow \{0, 1\}$  is a representation as an XOR-sum of conjunctions:

$$f(x_1, \dots, x_n) = \bigoplus_{S \subseteq [n]} a_S \cdot \prod_{i \in S} x_i,$$

where  $a_S \in \{0, 1\}$  and  $\oplus$  is XOR (addition mod 2).

The value of  $f$  on input set  $B$  is:

$$f(B) = \bigoplus_{A \subseteq B} a_A.$$

This is convolution with  $\zeta$  on the Boolean lattice, but over the field  $GF(2)$ .

By Möbius inversion over  $GF(2)$ , subtraction equals addition, so  $(-1)^{|B|-|A|} \equiv 1$ :

$$a_B = \bigoplus_{A \subseteq B} f(A).$$

## Zhegalkin Polynomials [2]

*Example:* Let  $f(x, y) = x$  and  $y$ . Truth table:  $f = (0, 0, 0, 1)$ .

Compute coefficients:

- $a_1 = f(00) = 0$ ,
- $a_x = f(00) \oplus f(10) = 0$ ,
- $a_y = f(00) \oplus f(01) = 0$ ,
- $a_{xy} = f(00) \oplus f(10) \oplus f(01) \oplus f(11) = 0 \oplus 0 \oplus 0 \oplus 1 = 1$ .

Result:  $f(x, y) = xy$ .

**Takeaway.** The standard algorithm for converting a truth table to Zhegalkin form (the “triangle method”) is exactly Möbius inversion on the Boolean cube over  $GF(2)$ .

## Counting Surjections

*Example:* How many *surjections*  $h : [k] \rightarrow [n]$  are there?

Define:

- $g(S) = |\{h : [k] \rightarrow [n] \mid \text{image of } h \text{ equals } S\}|$  for  $S \subseteq [n]$ .
- $F(S) = |\{h : [k] \rightarrow [n] \mid \text{image of } h \subseteq S\}| = |S|^k$ .

Then  $F(S) = \sum_{T \subseteq S} g(T)$ , i.e.,  $F = g * \zeta$ .

By Möbius inversion on the Boolean lattice:

$$\begin{aligned} g([n]) &= \sum_{T \subseteq [n]} (-1)^{n-|T|} F(T) \\ &= \sum_{T \subseteq [n]} (-1)^{n-|T|} |T|^k \\ &= \sum_{i=0}^n (-1)^{n-i} \binom{n}{i} i^k \end{aligned}$$

This is exactly the PIE formula for surjections that we derived earlier!

## Computing $\mu$ on the Divisor Lattice

Before stating the general formula, let us compute  $\mu$  on the divisor lattice of 12.

*Example:* Divisors of  $12 = 2^2 \cdot 3$  are:  $\{1, 2, 3, 4, 6, 12\}$ .

Using the recursion  $\mu(1, d) = - \sum_{\substack{c \mid d, \\ c < d}} \mu(1, c)$ :

| $d$ | $\mu(1, d)$ | Computation            |
|-----|-------------|------------------------|
| 1   | 1           | base case              |
| 2   | -1          | $-\mu(1, 1)$           |
| 3   | -1          | $-\mu(1, 1)$           |
| 4   | 0           | $-(1 + (-1))$          |
| 6   | 1           | $-(1 + (-1) + (-1))$   |
| 12  | 0           | $-(1 - 1 - 1 + 0 + 1)$ |

Notice:  $\mu(d) = 0$  when  $d$  has a squared prime factor ( $4 = 2^2$ ,  $12 = 2^2 \cdot 3$ ), and  $\mu(d) = (-1)^k$  when  $d$  is a product of  $k$  distinct primes.

# The Classical Möbius Function

Consider the divisor lattice  $\mathcal{P} = (\mathbb{Z}^+, |)$ .

**Theorem 39:** The Möbius function on  $(\mathbb{Z}^+, |)$  is the *classical Möbius function*:

$$\mu(d) = \begin{cases} 1 & \text{if } d = 1 \\ (-1)^k & \text{if } d = p_1 \cdots p_k \text{ (distinct primes)} \\ 0 & \text{if } d \text{ has a squared prime factor} \end{cases}$$

where  $\mu(a, b) = \mu(b/a)$  for  $a \mid b$ .

**Proof:** We need  $\sum_{d \mid n} \mu(d) = \begin{cases} 1 & \text{if } n=1 \\ 0 & \text{if } n>1 \end{cases}$ .

**Case  $n = 1$ :**  $\sum_{d \mid 1} \mu(d) = \mu(1) = 1$ . ✓

**Case  $n > 1$ :** Let  $n = p_1^{a_1} \cdots p_r^{a_r}$ . Divisors with a squared factor have  $\mu(d) = 0$ , so only *square-free* divisors contribute:

## The Classical Möbius Function [2]

$$\begin{aligned}\sum_{d \mid n} \mu(d) &= \sum_{S \subseteq [r]} \mu\left(\prod_{i \in S} p_i\right) \\ &= \sum_{S \subseteq [r]} (-1)^{|S|} \\ &= \sum_{j=0}^r \binom{r}{j} (-1)^j \\ &= (1 - 1)^r = 0\end{aligned}$$

□

The identity  $\mu(a, b) = \mu(b/a)$  holds because the interval  $[a, b]$  in the divisor lattice is isomorphic to the divisor lattice of  $b/a$ .

*Example:*

- $\mu(1) = 1, \mu(2) = -1, \mu(3) = -1$
- $\mu(4) = 0$  (since  $4 = 2^2$ )
- $\mu(6) = \mu(2 \cdot 3) = 1, \mu(30) = \mu(2 \cdot 3 \cdot 5) = -1$

## Number-Theoretic Möbius Inversion

**Theorem 40:** For arithmetic functions  $f, g : \mathbb{Z}^+ \rightarrow R$ :

$$f(n) = \sum_{d \mid n} g(d) \iff g(n) = \sum_{d \mid n} \mu(d) \cdot f(n/d)$$

This is exactly Möbius inversion on  $(\mathbb{Z}^+, |)$ , where the order relation is divisibility.

The convolution here is the classical *Dirichlet convolution* of arithmetic functions.

**Proof:** Suppose  $f(n) = \sum_{d \mid n} g(d)$ . This is the Dirichlet convolution  $f = g * \zeta$ . Convolving both sides with  $\mu$ :  $f * \mu = g * \zeta * \mu = g * \delta = g$ . Writing out explicitly:  $g(n) = \sum_{d \mid n} \mu(d) f(n/d)$ .  $\square$   $\square$

## Euler's Totient Function

*Example:* Define  $\varphi(n) = |\{k \in [n] \mid \gcd(k, n) = 1\}|$ .

Every  $k \in [n]$  has  $\gcd(k, n) = d$  for a unique  $d \mid n$ , so  $n = \sum_{d \mid n} \varphi(d)$ .

By Möbius inversion:  $\varphi(n) = n \cdot \sum_{d \mid n} \frac{\mu(d)}{d}$ .

Only square-free divisors survive, so the sum simplifies:

$$\varphi(n) = n \cdot \prod_{\substack{p \mid n, \\ p \text{ prime}}} \left(1 - \frac{1}{p}\right)$$

*Example:* For  $n = 12 = 2^2 \cdot 3$ , we have:

$$\varphi(12) = 12 \cdot \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{3}\right) = 12 \cdot \frac{1}{2} \cdot \frac{2}{3} = 4$$

The four numbers coprime to 12 are 1, 5, 7, 11. ✓

## Summary: Möbius Inversion

We saw four posets, each with its own Möbius function:

| Poset                          | Möbius function                   | What it gives                              |
|--------------------------------|-----------------------------------|--------------------------------------------|
| Chain $(\mathbb{N}, \leq)$     | $\mu(n, n) = 1, \mu(n-1, n) = -1$ | Finite differences: $F(n) = G(n) - G(n-1)$ |
| Boolean $(2^{[n]}, \subseteq)$ | $\mu(A, B) = (-1)^{ B - A }$      | Inclusion-exclusion                        |
| Boolean over $GF(2)$           | $\mu(A, B) = 1$ always            | Zhegalkin polynomials                      |
| Divisors $(\mathbb{Z}^+,  )$   | $\mu(d)$ : classical              | Number-theoretic inversion                 |

**Common pattern:**  $\zeta$  sums over everything below,  $\mu$  undoes the sum.

The poset determines the form of  $\mu$ .

# The Twelfold Way

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*“The purpose of computing is insight, not numbers.”*

*— Richard Hamming*

## Balls and Boxes

Consider the most basic combinatorial question:

**How many ways can we place  $n$  balls into  $k$  boxes?**

There are three independent dimensions of variation:

### Balls

1. Distinct (labeled  $1, \dots, n$ )
2. Indistinguishable (identical)

### Boxes

1. Distinct (labeled  $1, \dots, k$ )
2. Indistinguishable (unlabeled)

### Placement

1. Unrestricted
2. Injective ( $\leq 1$  per box)
3. Surjective ( $\geq 1$  per box)

Total:  $2 \times 2 \times 3 = 12$  cases.

## Small Examples

*Example:* Place  $n = 2$  distinct balls into  $k = 2$  distinct boxes, surjectively.

| Placement                                              | Description              |
|--------------------------------------------------------|--------------------------|
| Ball 1 $\rightarrow$ Box 1, Ball 2 $\rightarrow$ Box 2 | each ball in its own box |
| Ball 1 $\rightarrow$ Box 2, Ball 2 $\rightarrow$ Box 1 | balls swapped            |

Answer:  $k!s_k^{\text{II}}(n) = 2! \cdot s_2^{\text{II}}(2) = 2 \cdot 1 = 2$ . ✓

*Example:* Place  $n = 3$  indistinguishable balls into  $k = 2$  distinct boxes, surjectively.

Each box gets  $\geq 1$ , so we need compositions of 3 into 2 positive parts:  $3 = 1 + 2 = 2 + 1$ .

Answer:  $\binom{n-1}{k-1} = \binom{2}{1} = 2$ . ✓

## Distinct Balls, Distinct Boxes

This is the theory of *functions*  $f : [n] \rightarrow [k]$ .

### Unrestricted

Any function:  $k^n$ .

Each of  $n$  balls has  $k$  choices.

Multiplication principle.

### Injective

At most one ball per box:

$$(k)_n = k! / (k - n)!.$$

The number of injective functions  $[n] \rightarrow [k]$ . Requires  $k \geq n$ .

### Surjective

Every box gets  $\geq 1$  ball:

$$k! s_k^{\text{II}}(n).$$

Partition  $[n]$  into  $k$  labeled blocks (Stirling number  $s_k^{\text{II}}(n)$ ), then assign the  $k$  labels to blocks in  $k!$  ways.

The surjection count uses PIE (equivalently, Möbius inversion on the Boolean lattice):

$$k! s_k^{\text{II}}(n) = \sum_{i=0}^k (-1)^i \binom{k}{i} (k - i)^n.$$

# Indistinguishable Balls, Distinct Boxes

This is the theory of *multisets* and *compositions*.

## Unrestricted

Multiset of size  $n$  from  $[k]$ :

$$\binom{n+k-1}{n}.$$

*Stars and bars*: arrange  $n$  stars and  $k - 1$  bars.

## Injective

At most one per box:  $\binom{k}{n}$ .

Choose which  $n$  of  $k$  boxes are occupied. Requires  $k \geq n$ .

## Surjective

Every box gets  $\geq 1$ :  $\binom{n-1}{k-1}$ .

*Compositions* of  $n$  into  $k$  positive parts (give each box 1 first, then distribute  $n - k$  freely). Requires  $n \geq k$ .

## Distinct Balls, Indistinguishable Boxes

This is the theory of *set partitions*.

### Unrestricted

Into  $\leq k$  non-empty blocks:

$$\sum_{j=1}^k s_j^{\text{II}}(n).$$

Sum of Stirling numbers. If  $k \geq n$ , this equals  $B_n$ .

### Injective

At most one ball per box:

$$[n \leq k].$$

If  $n \leq k$ : each ball in its own box, one way. If  $n > k$ : impossible (0).

### Surjective

Into exactly  $k$  non-empty blocks:  $s_k^{\text{II}}(n)$ .

Stirling number of the second kind. Requires  $n \geq k$ .

Here  $[P]$  denotes the *Iverson bracket*:

$[P] = 1$  if  $P$  is true, 0 otherwise.

# Indistinguishable Balls, Indistinguishable Boxes

This is the theory of *integer partitions*.

## Unrestricted

Into  $\leq k$  parts:  $\sum_{j=1}^k p_{j(n)}$ .

Sum of partition counts into exactly  $j$  parts.

## Injective

At most one ball per box:  
 $[n \leq k]$ .

Same as Row 3: if  $n \leq k$ ,  
place one per box. If  $n > k$ :  
impossible.

## Surjective

Into exactly  $k$  parts:  $p_{k(n)}$ .

Integer partitions of  $n$  into  
exactly  $k$  parts. Requires  $n \geq k$ .

Here  $p_{j(n)}$  denotes the number of integer partitions of  $n$  into exactly  $j$  parts.

Note that Rows 3 and 4 are *unlabeled*, so the formulas differ from Rows 1 and 2 only in “forgetting” the box labels.

## From Unrestricted to Surjective

The surjective column is obtained from the unrestricted column by Möbius inversion on the *Boolean lattice*.

### Distinct balls, distinct boxes:

Unrestricted:  $k^n$  (all functions).

Surjective:  $k!s_k^{\text{II}}(n) = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$ .

This is PIE — Möbius inversion on  $2^{[k]}$ !

### Indistinguishable balls, distinct boxes:

Unrestricted:  $\binom{n+k-1}{n}$ .

Surjective:  $\binom{n-1}{k-1}$ .

The same PIE argument, applied to “is box  $i$  empty?” properties.

## Forgetting Labels

The transition from distinct to indistinguishable boxes is “forgetting the labels.” For *surjective* arrangements, every placement becomes one of  $k!$  equivalent arrangements (one for each permutation of the labels), so dividing by  $k!$  gives the count for indistinguishable boxes.

**Key identity:** dividing Row 1 surjective by  $k!$  gives Row 3 surjective:

$$s_k^\Pi(n) = \frac{1}{k!} \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$$

More generally, this “forget labels” operation is governed by the Möbius function of the *partition lattice*  $\Pi_k$ .

## The Twelfold Way

| Balls           | Boxes           | Unrestricted                | Injective      | Surjective         |
|-----------------|-----------------|-----------------------------|----------------|--------------------|
| <b>Distinct</b> | <b>Distinct</b> | $k^n$                       | $(k)_n$        | $k!s_k^{\Pi}(n)$   |
| <b>Indist.</b>  | <b>Distinct</b> | $\binom{n+k-1}{n}$          | $\binom{k}{n}$ | $\binom{n-1}{k-1}$ |
| <b>Distinct</b> | <b>Indist.</b>  | $\sum_{j=1}^k s_j^{\Pi}(n)$ | $[n \leq k]$   | $s_k^{\Pi}(n)$     |
| <b>Indist.</b>  | <b>Indist.</b>  | $\sum_{j=1}^k p_{j(n)}$     | $[n \leq k]$   | $p_{k(n)}$         |

All formulas hold for the standard non-degenerate cases  $n, k \geq 1$ .

# Course Wrap-Up

## Counting fundamentals

- Addition, multiplication, bijection principles
- Permutations and combinations
- Multisets and compositions
- Binomial and multinomial theorems

## Analytic methods

- Generating functions
- Characteristic equations
- Annihilators
- Master theorem, Akra–Bazzi method

## Structures

- Set partitions (Stirling, Bell)
- Integer partitions (Ferrers, Young tableaux)
- Inclusion–exclusion (PIE, derangements)

## Advanced topics

- Gamma function
- Möbius inversion
- The Twelfold Way

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